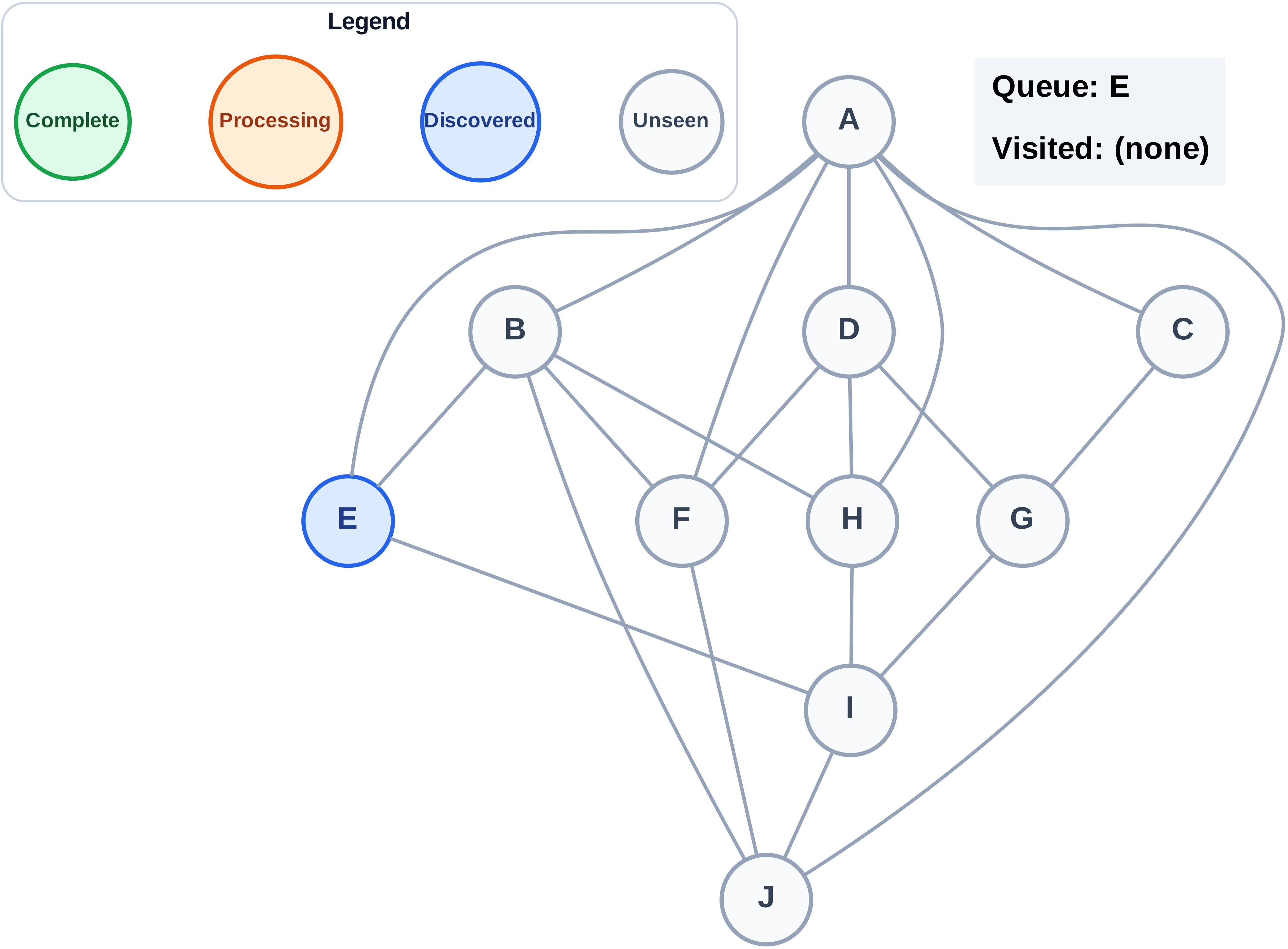
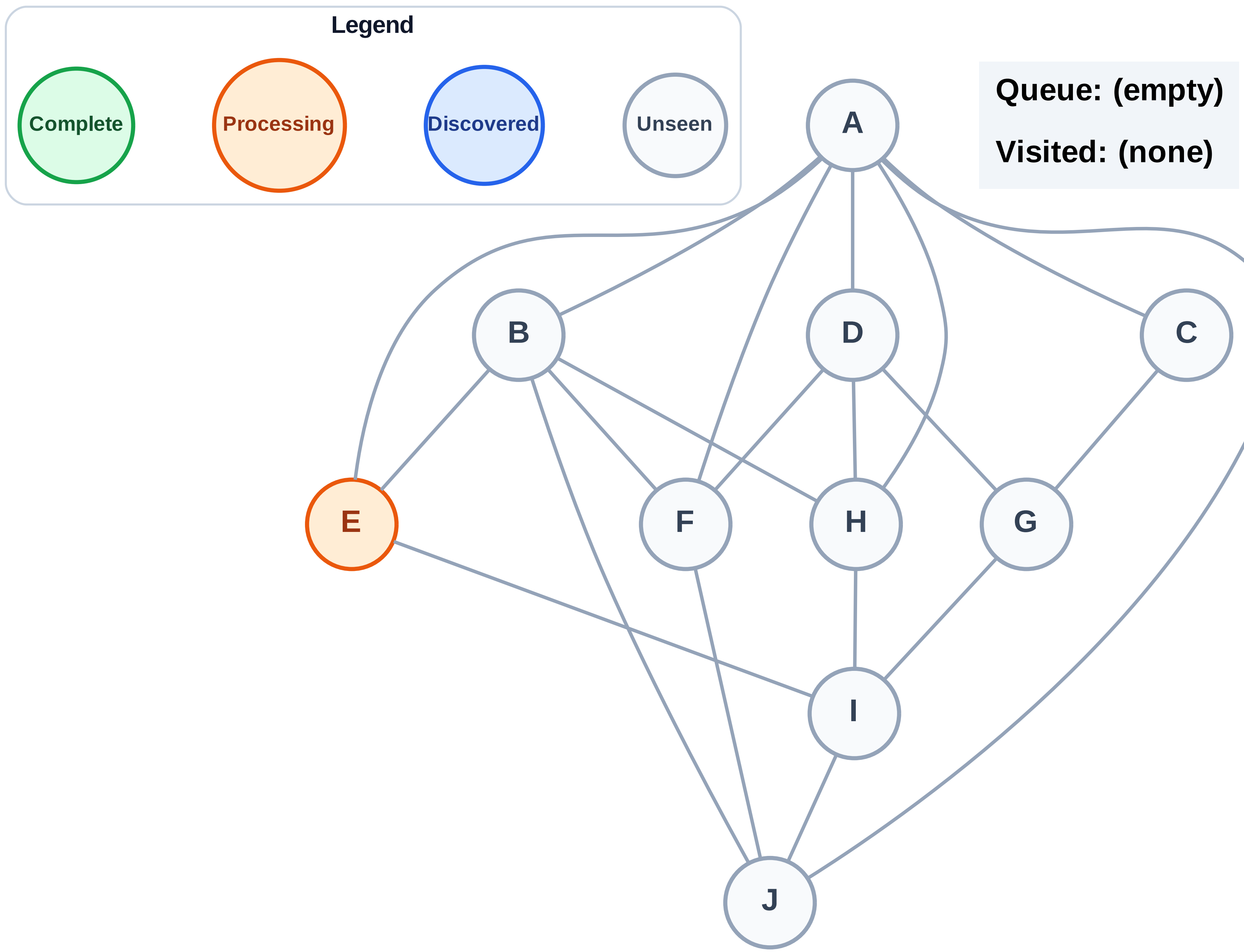
BFS Traversal

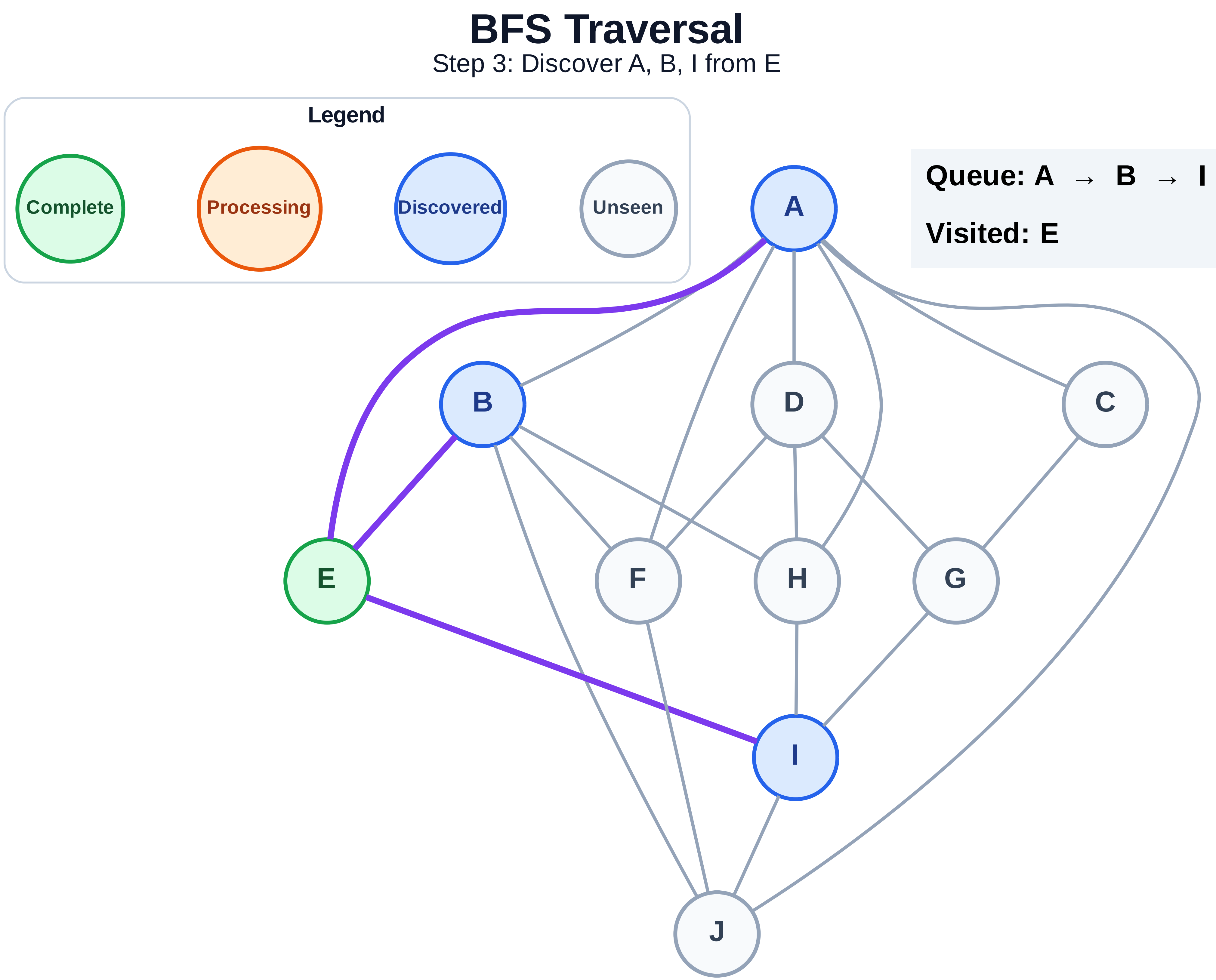
Step 1: Start at E



BFS Traversal

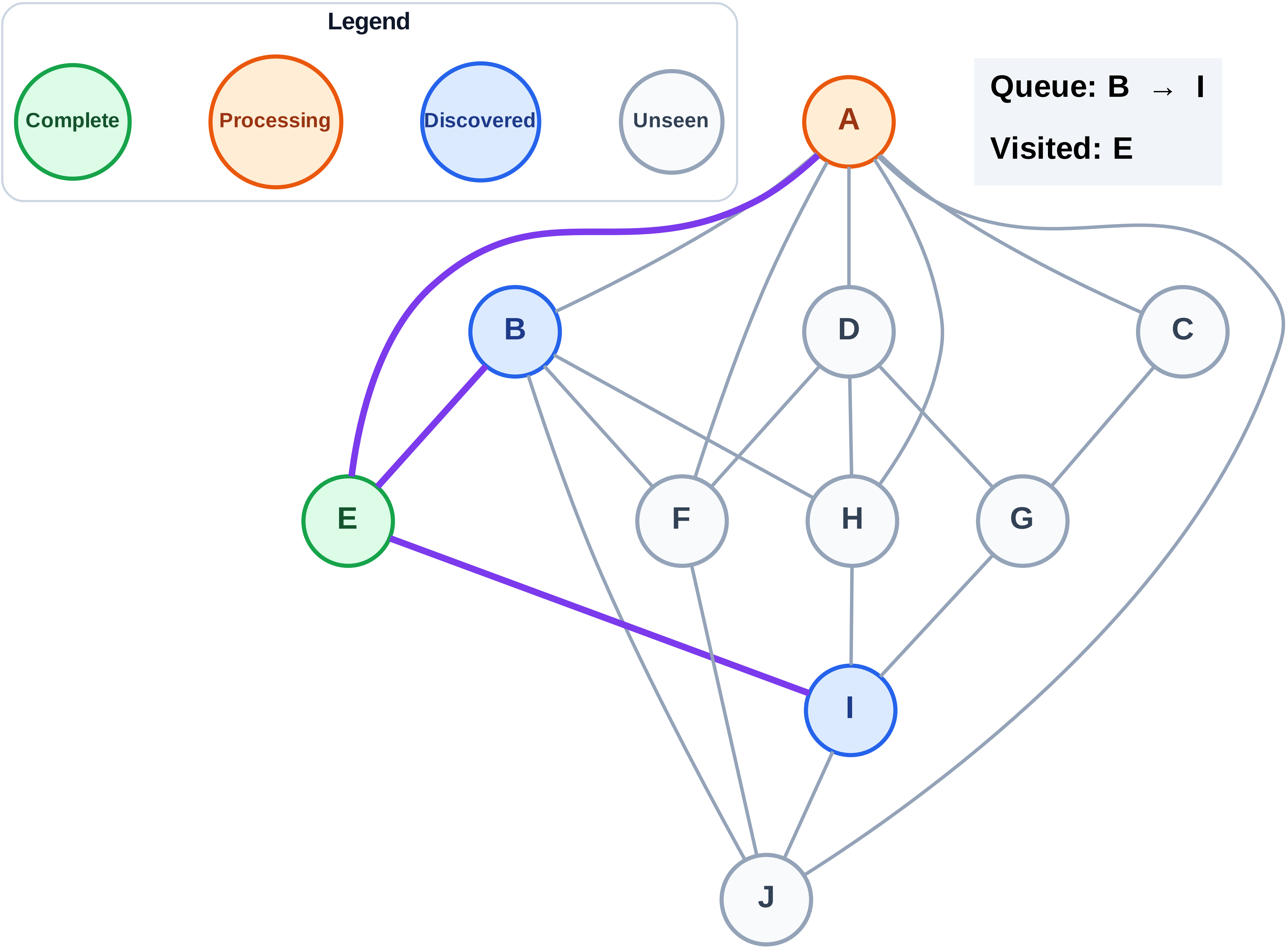
Step 2: Process node E




BFS Traversal
Step 3: Discover A, B, I from E
Legend
Complete
Processing
Discovered
Unseen
Queue: A → B → I
Visited: E
A
B
D
C
E
F
H
G
I
J

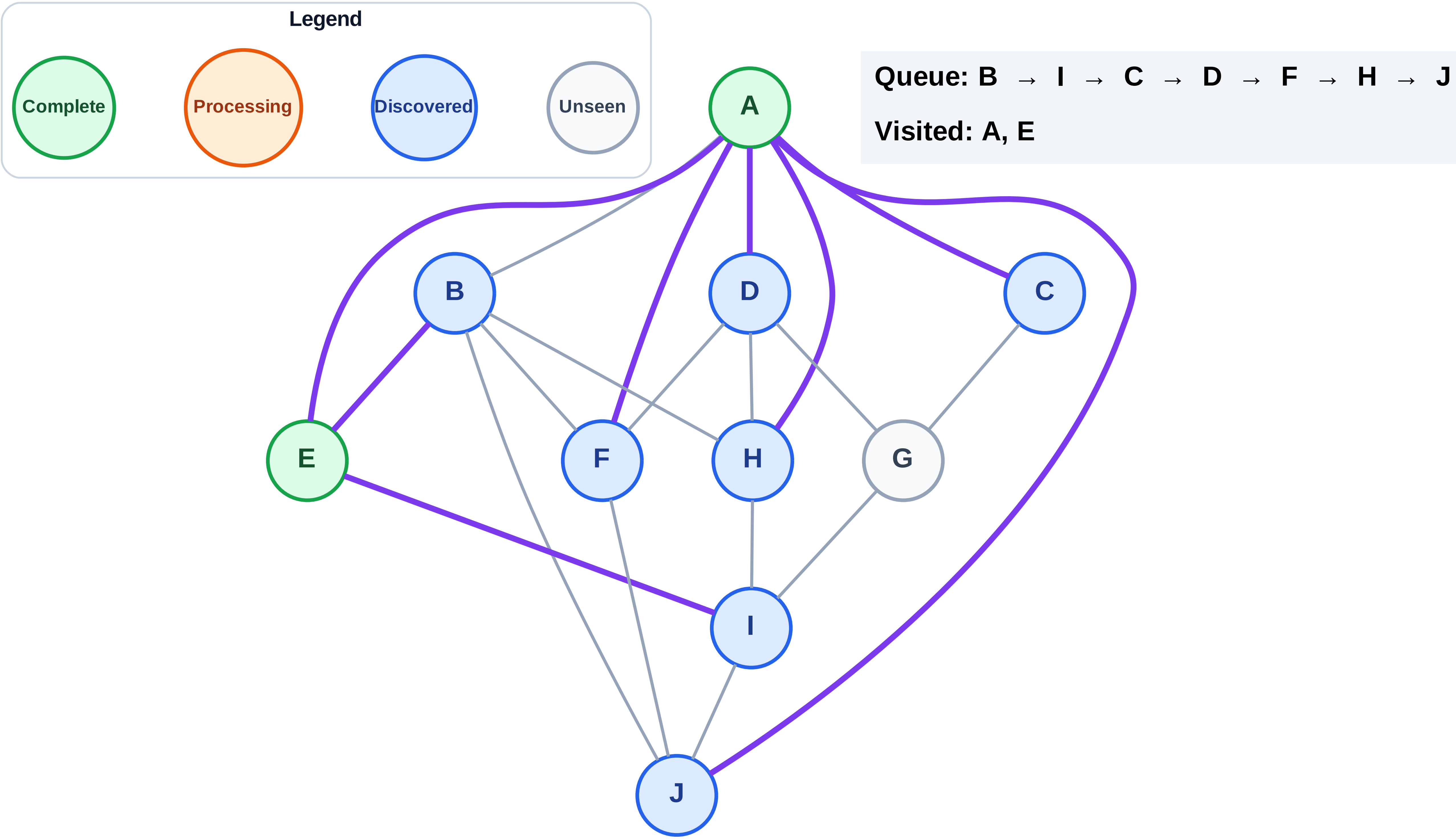
BFS Traversal

Step 4: Process node A



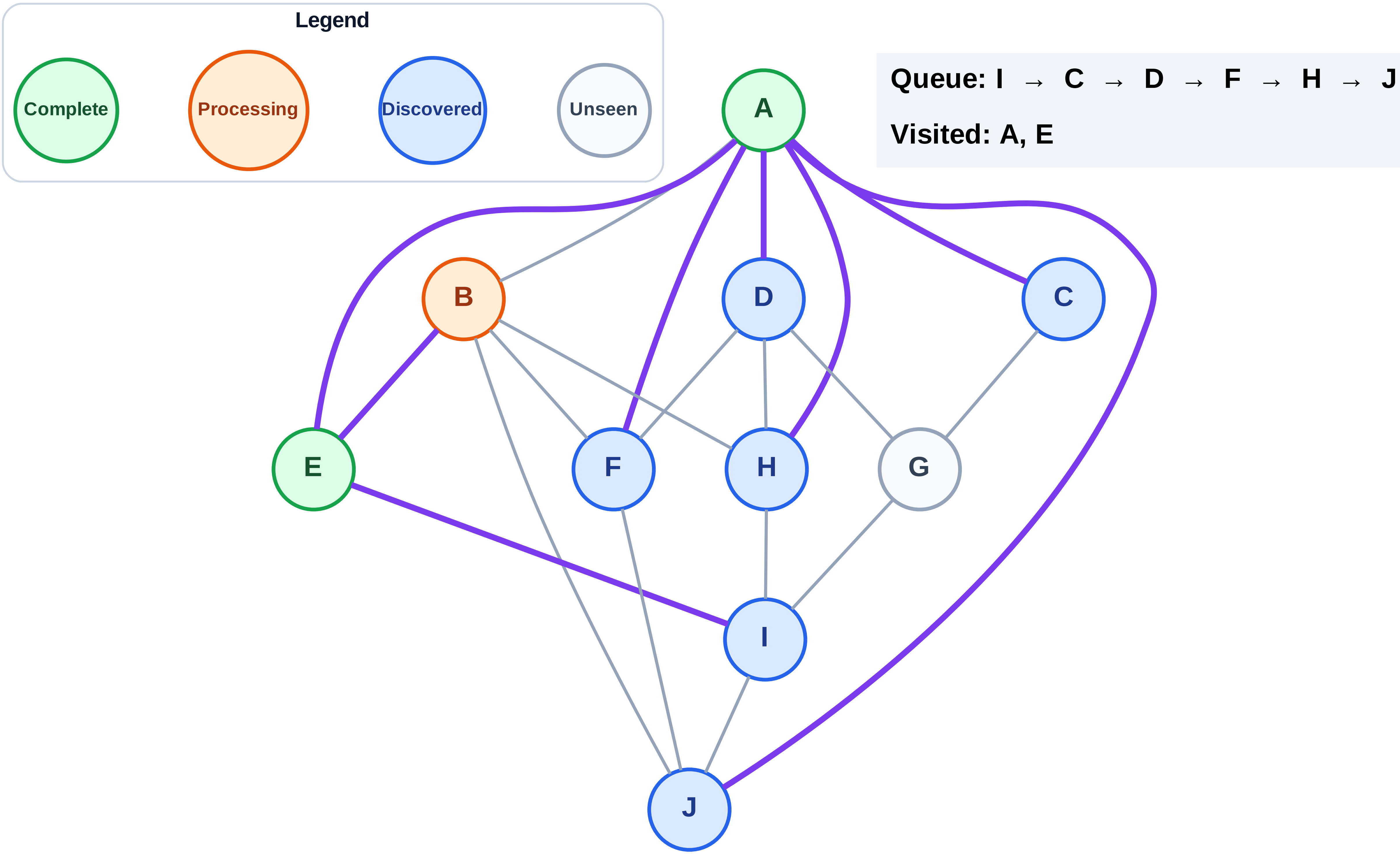
BFS Traversal

Step 5: Discover C, D, F, H, J from A



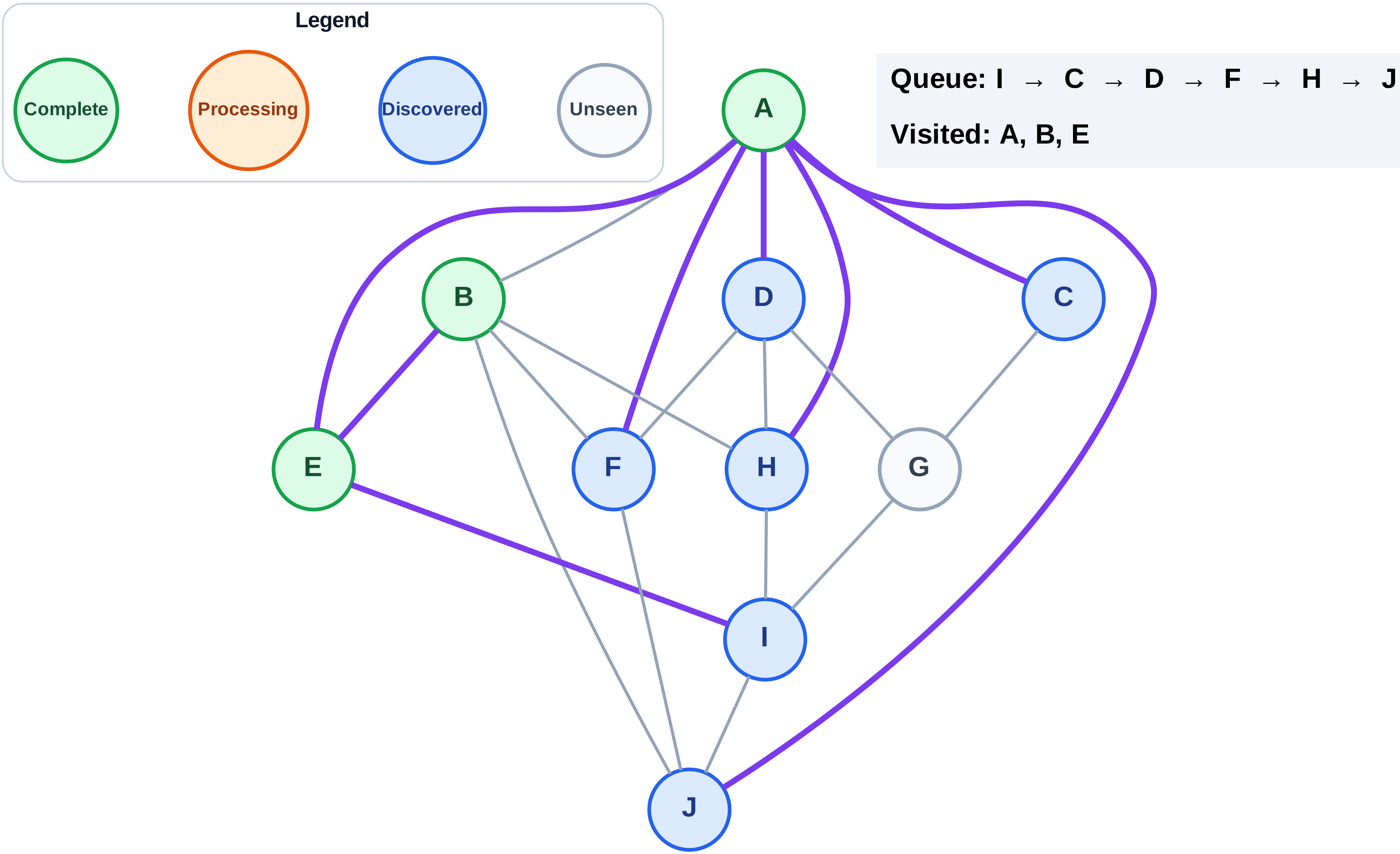
BFS Traversal

Step 6: Process node B



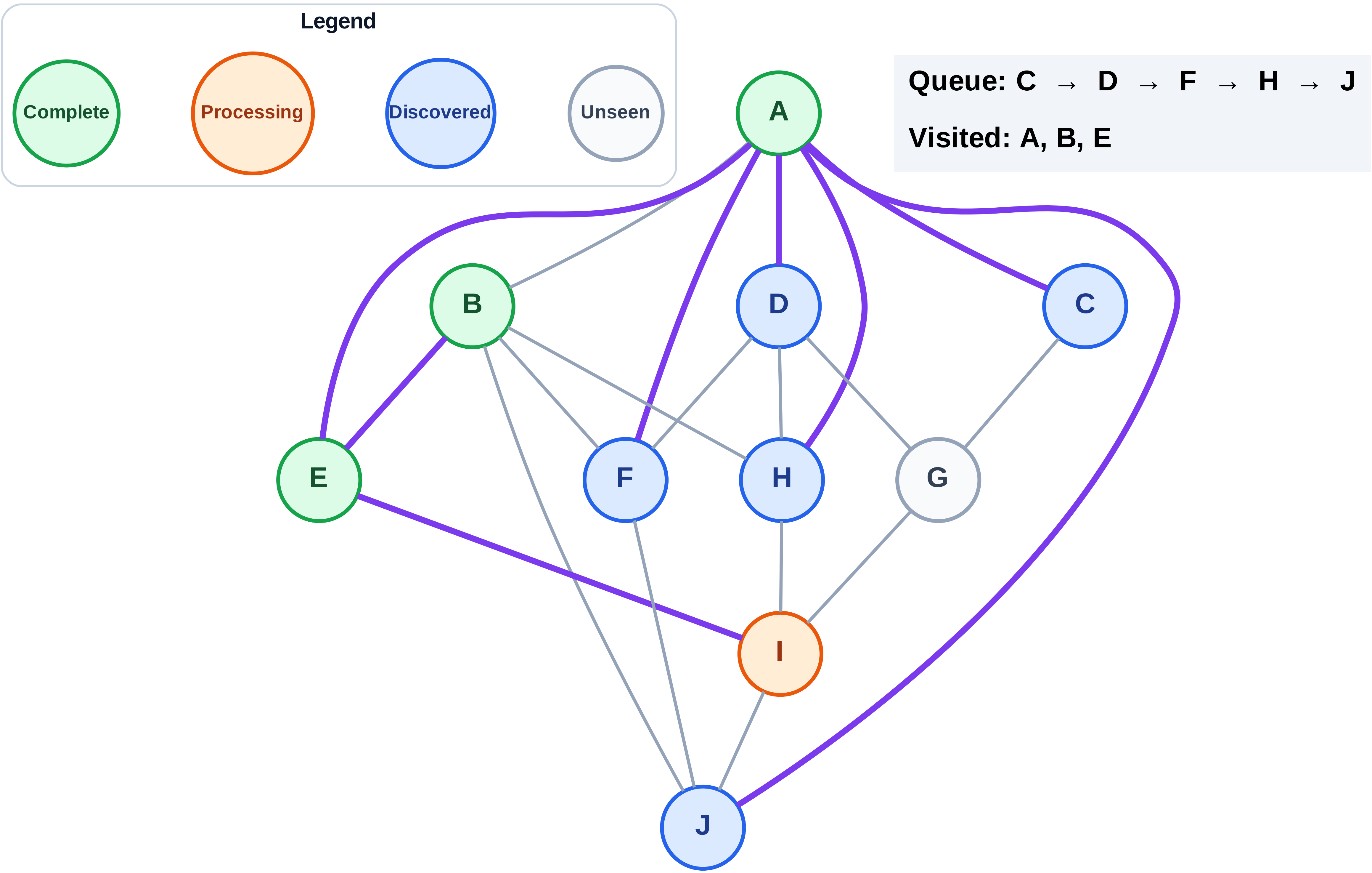
BFS Traversal

Step 7: No new neighbors from B



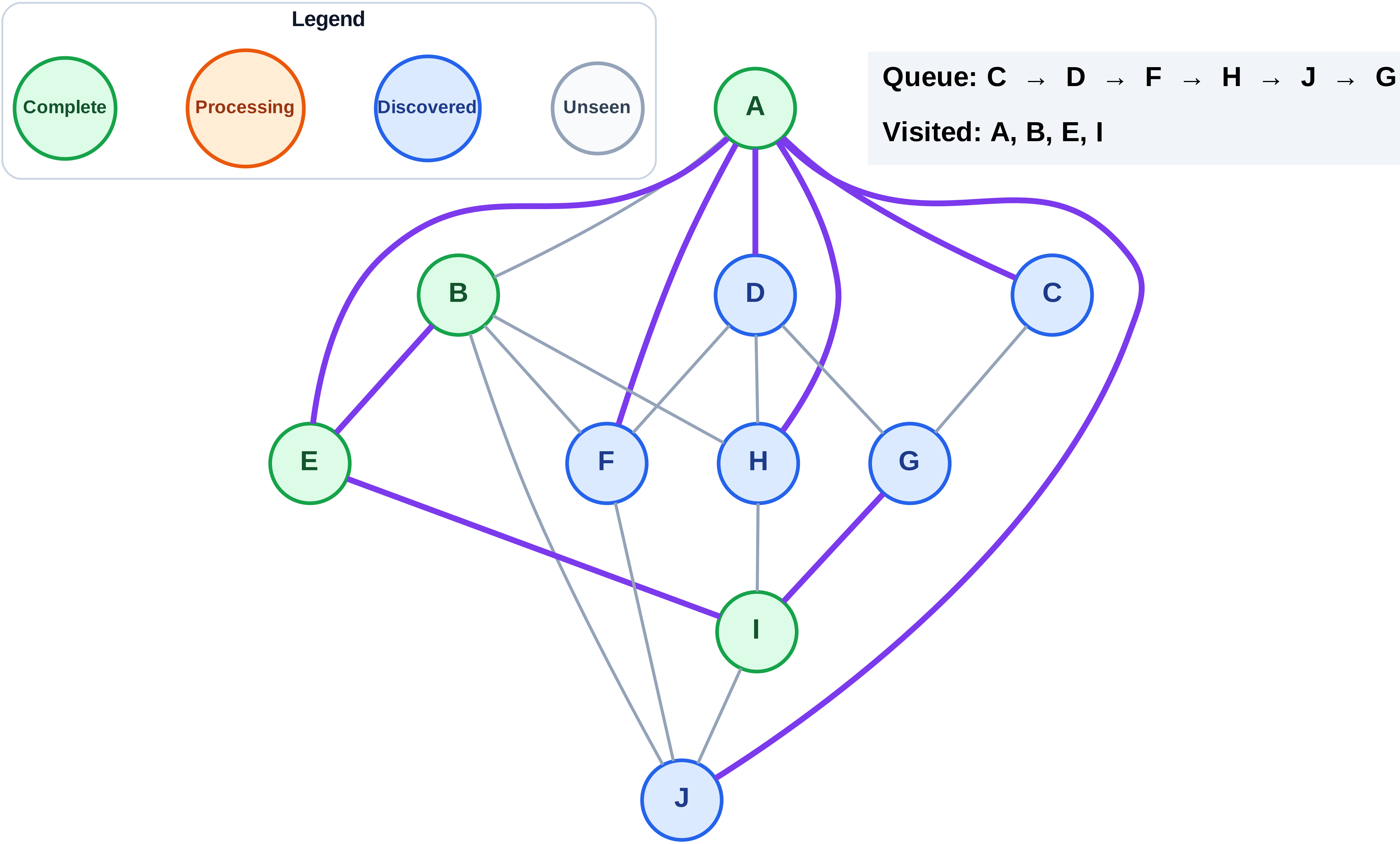
BFS Traversal

Step 8: Process node I



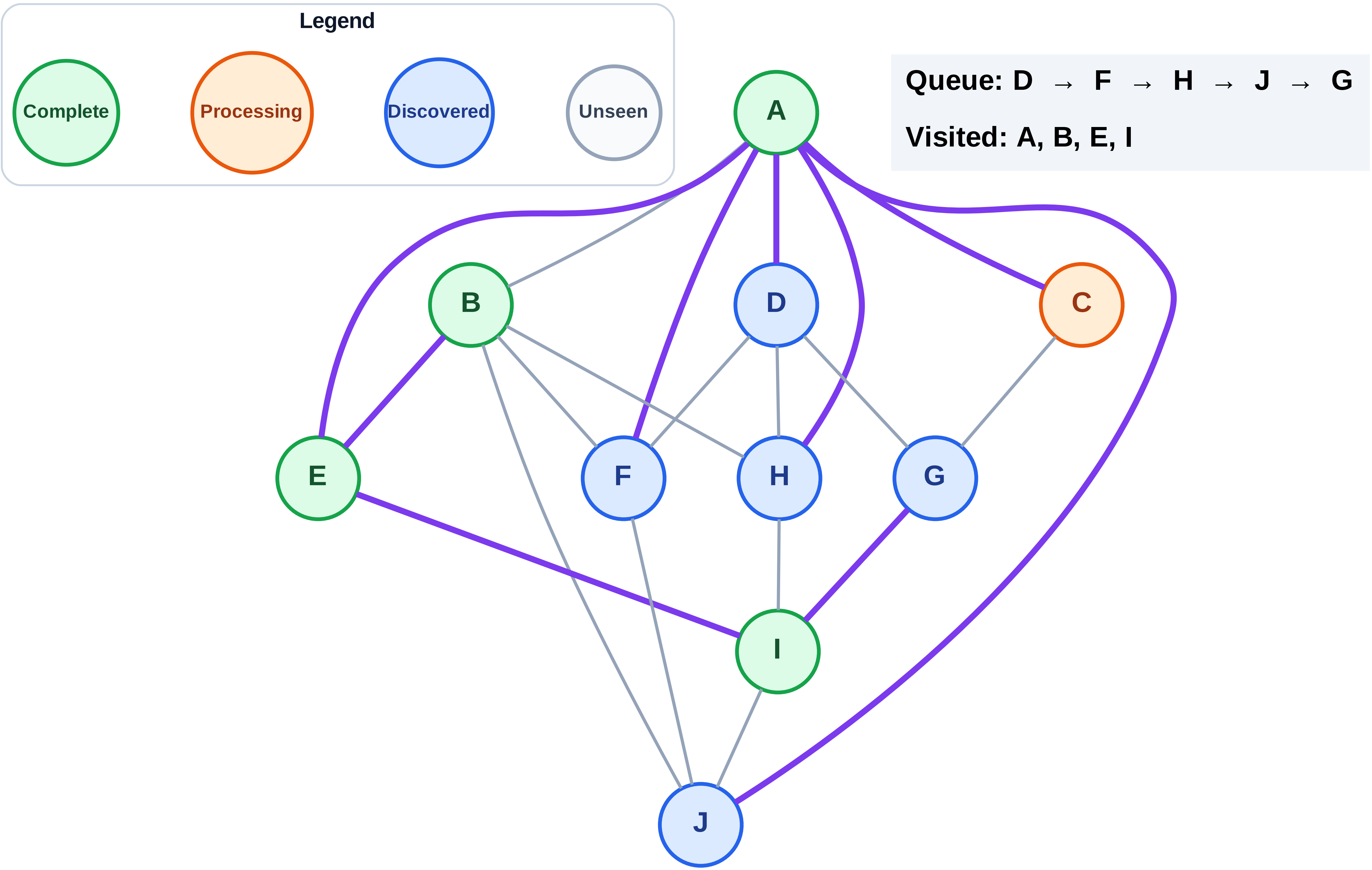
BFS Traversal

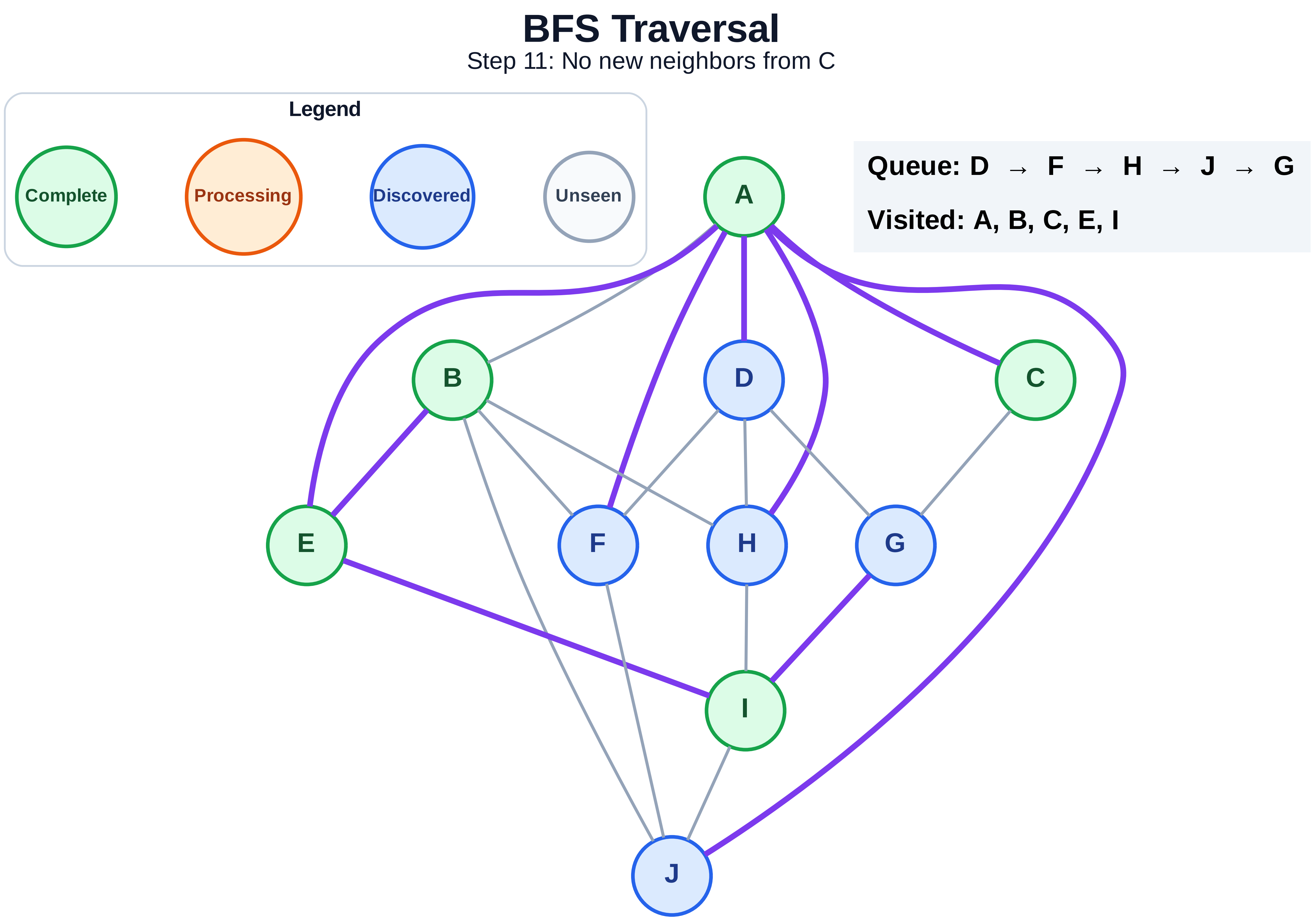
Step 9: Discover G from I



BFS Traversal

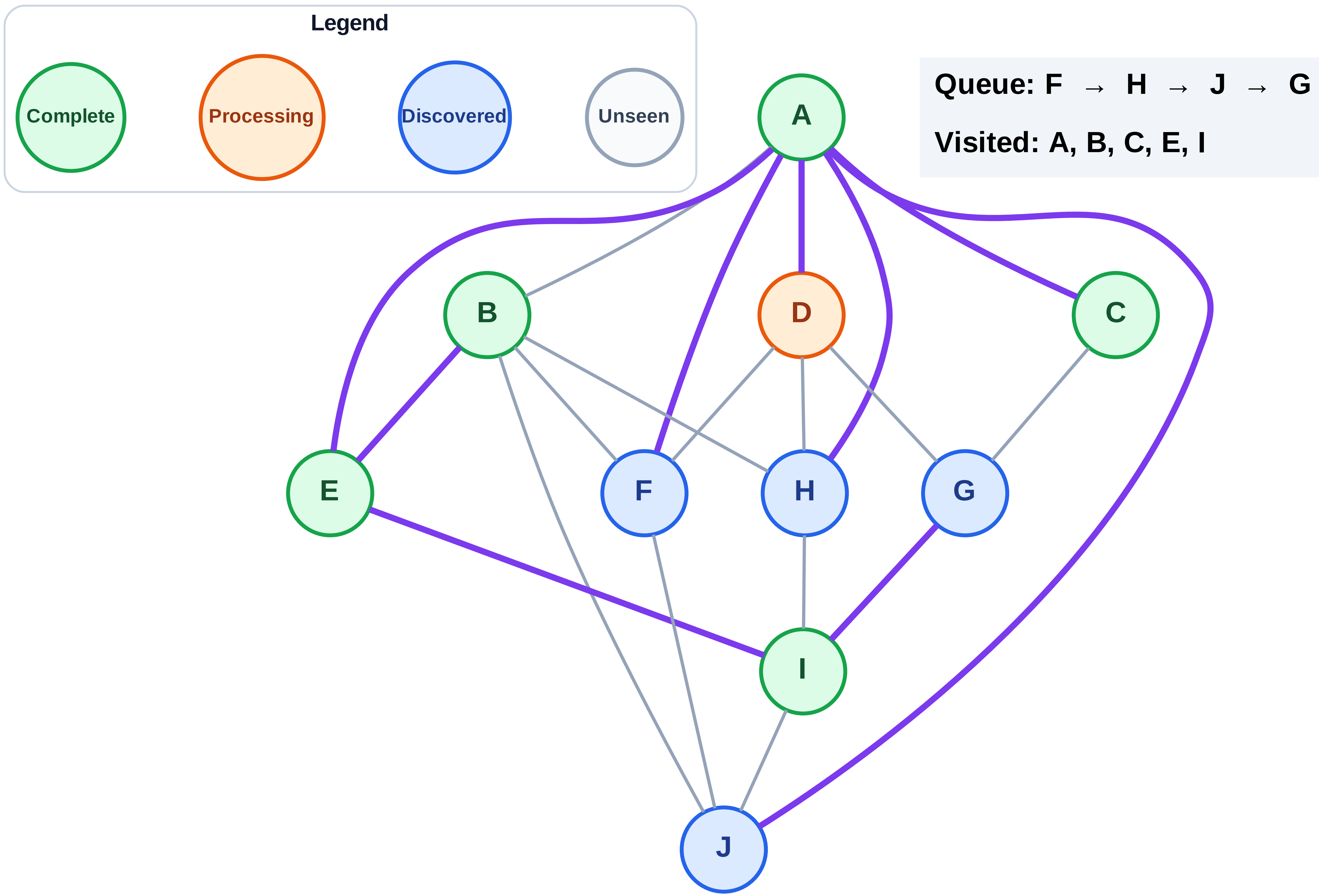
Step 10: Process node C




BFS Traversal
Step 11: No new neighbors from C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → F → H → J → G
Visited: A, B, C, E, I
A
B
D
C
E
F
H
G
I
J

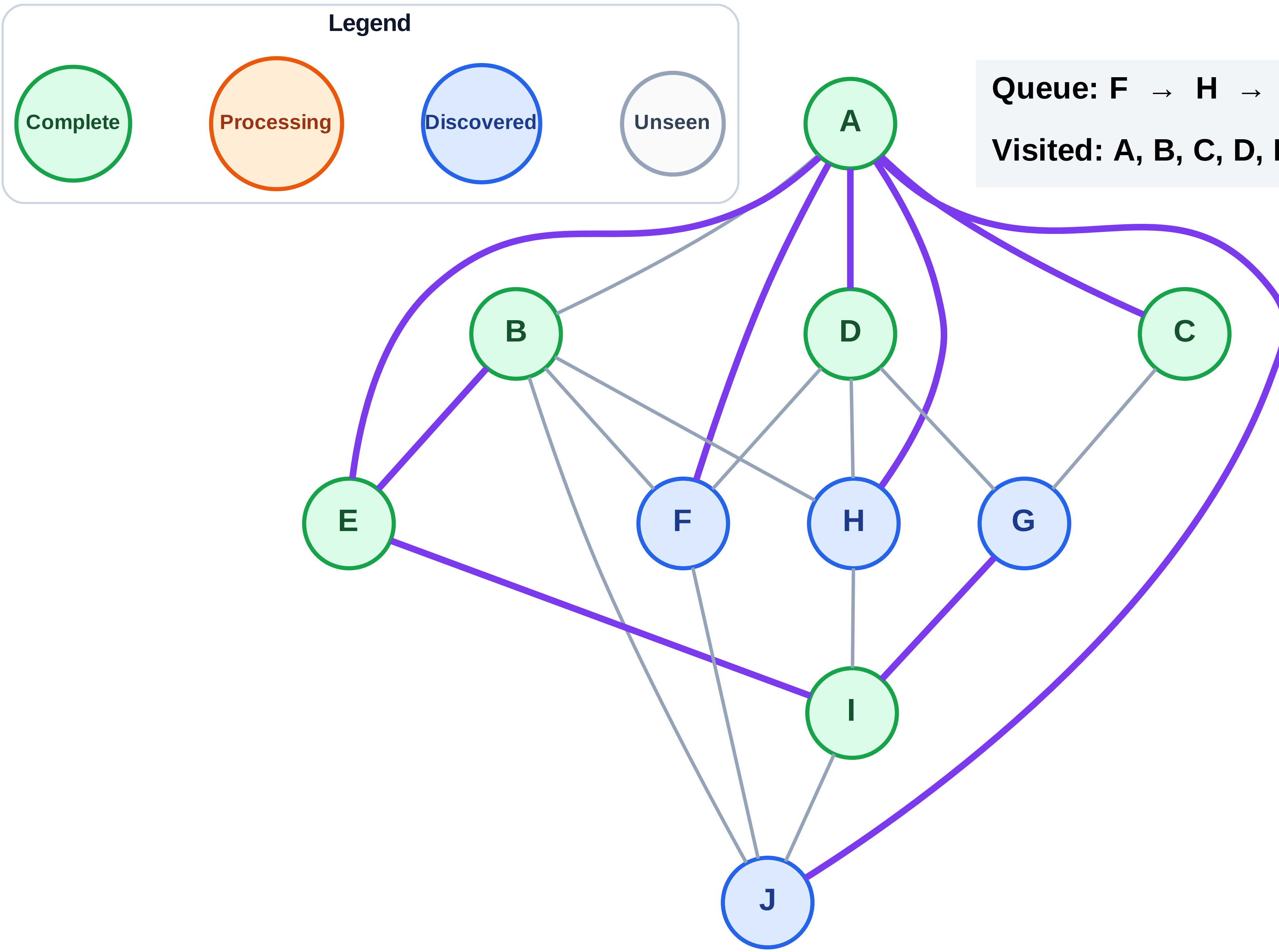
BFS Traversal

Step 12: Process node D



BFS Traversal

Step 13: No new neighbors from D

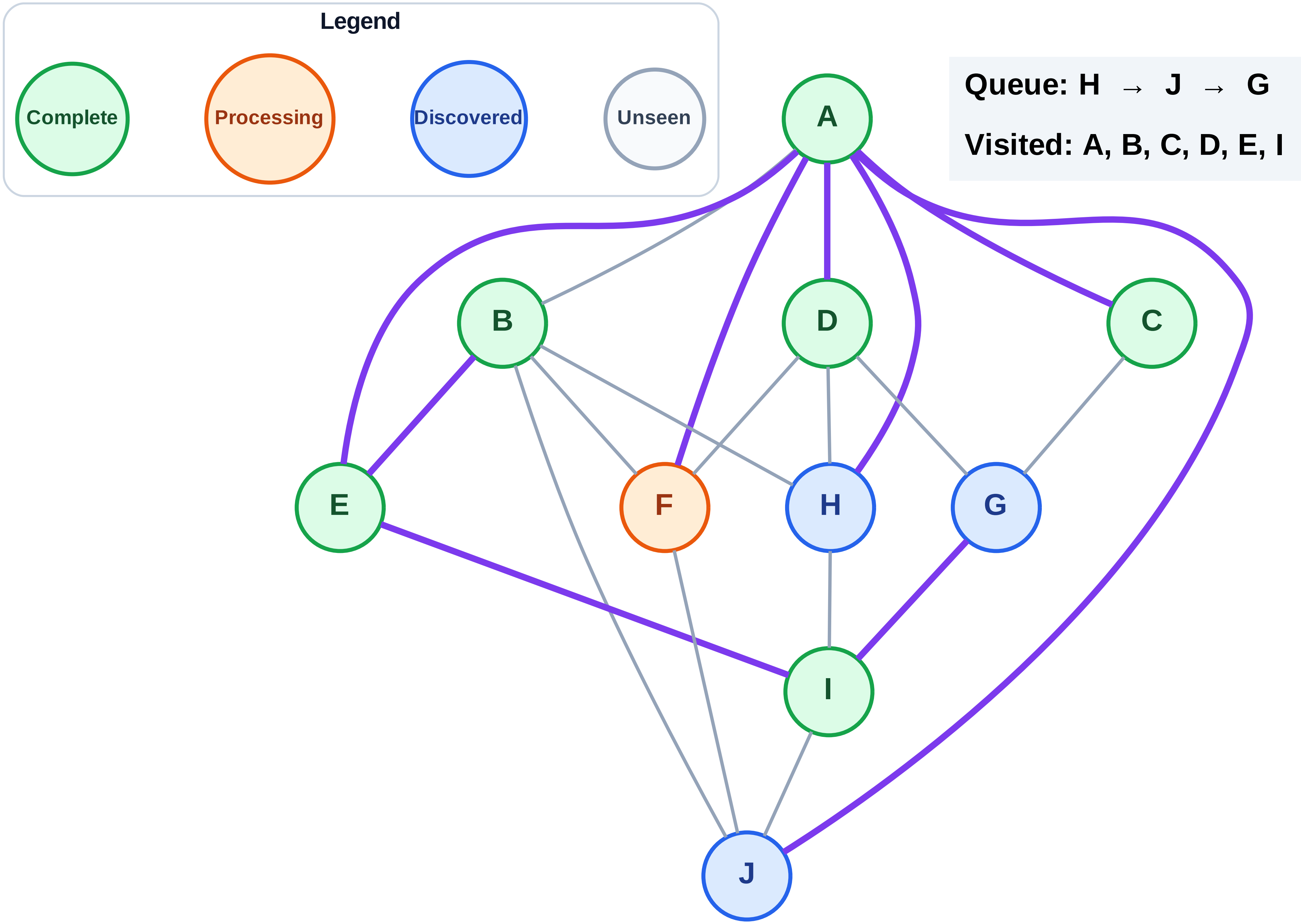


Queue: F → H → J → G

Visited: A, B, C, D, E, I

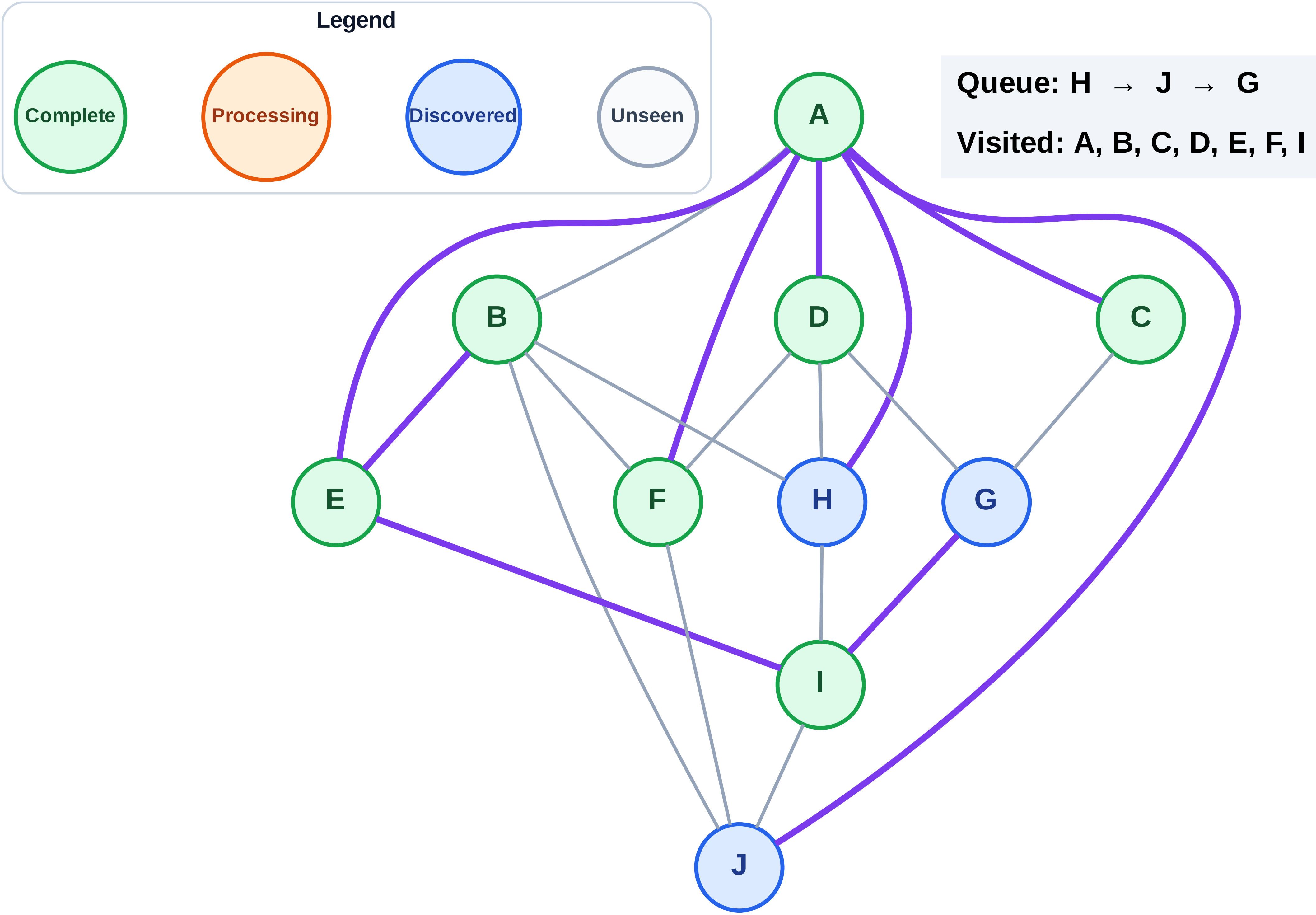
BFS Traversal

Step 14: Process node F



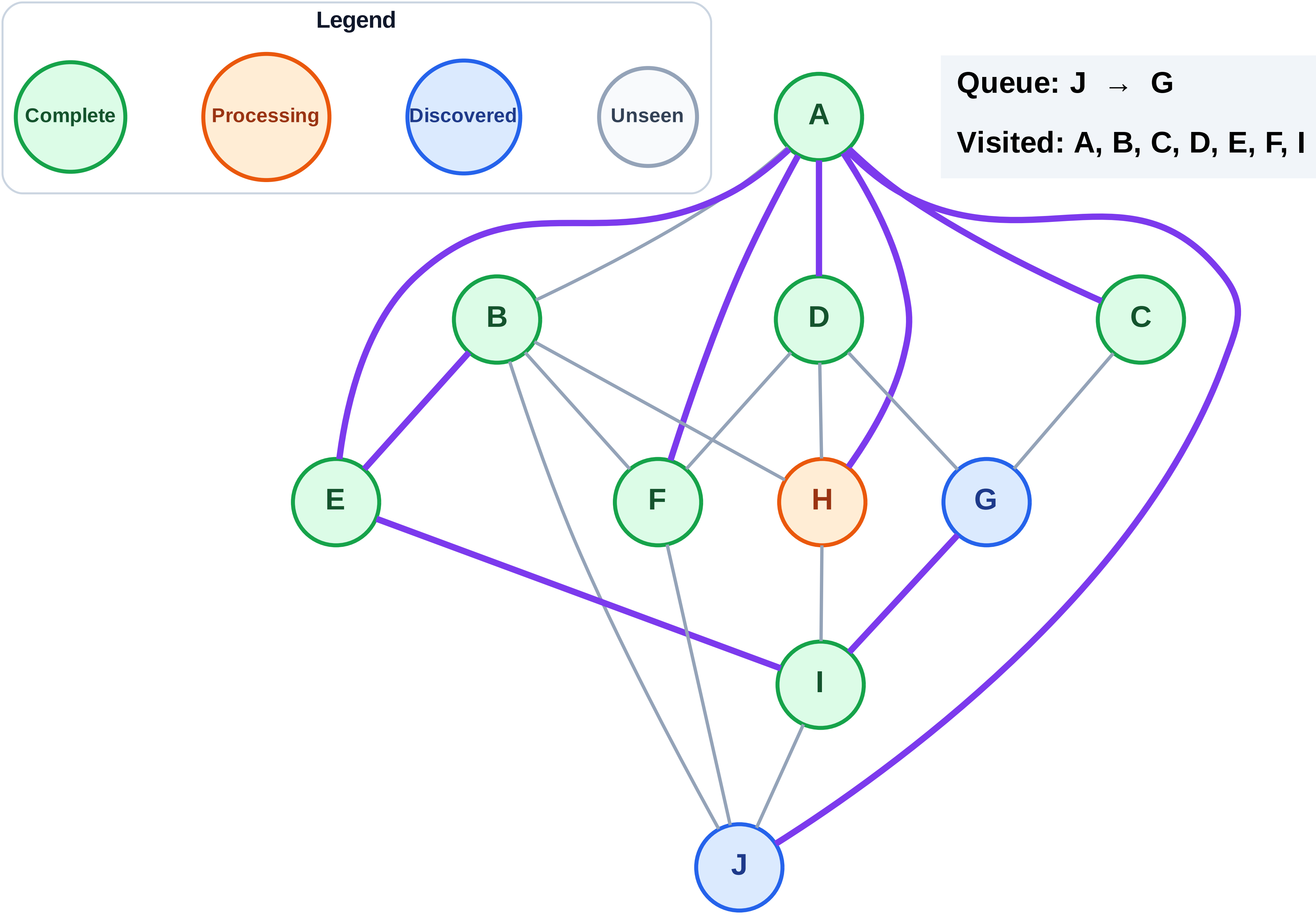
BFS Traversal

Step 15: No new neighbors from F



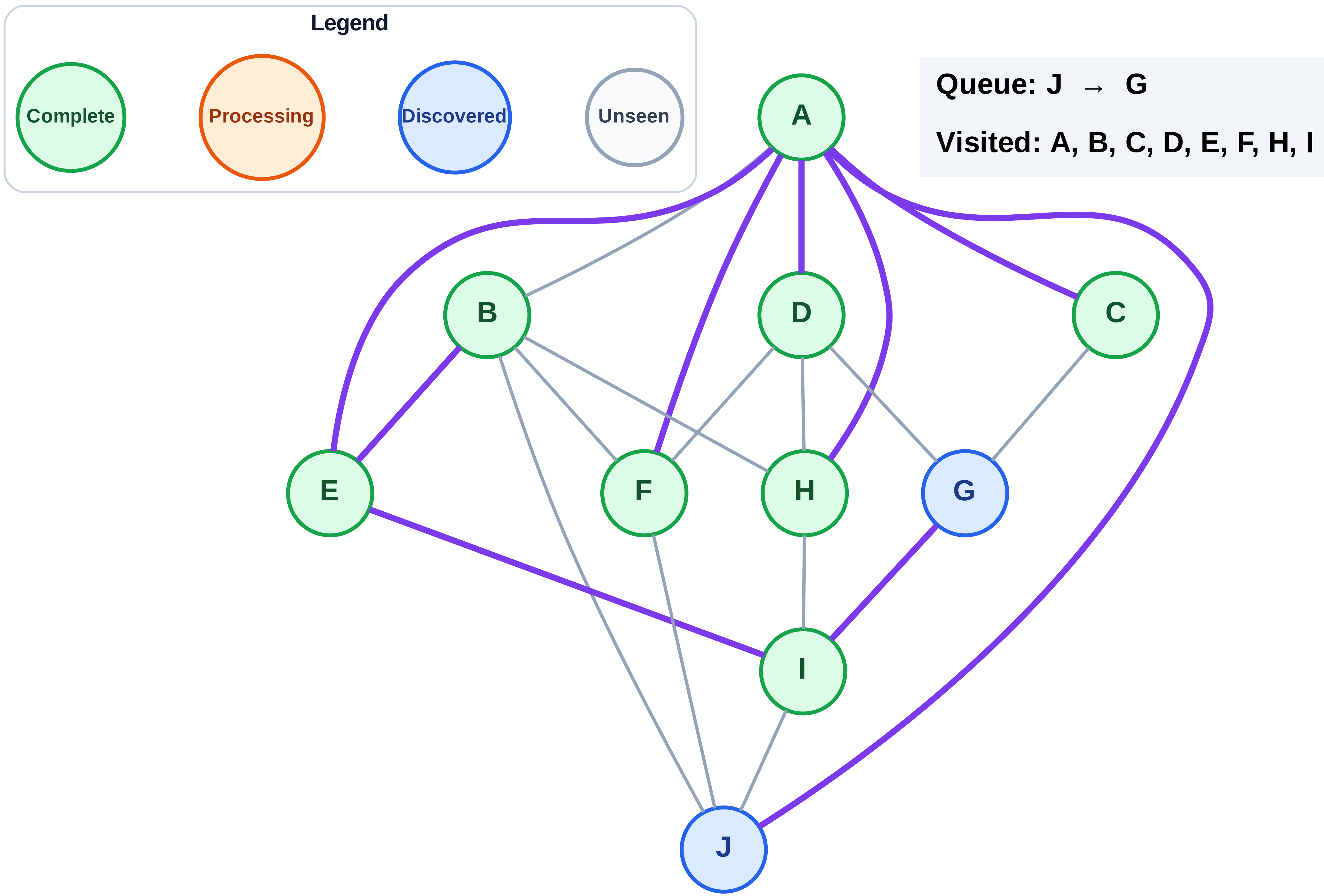
BFS Traversal

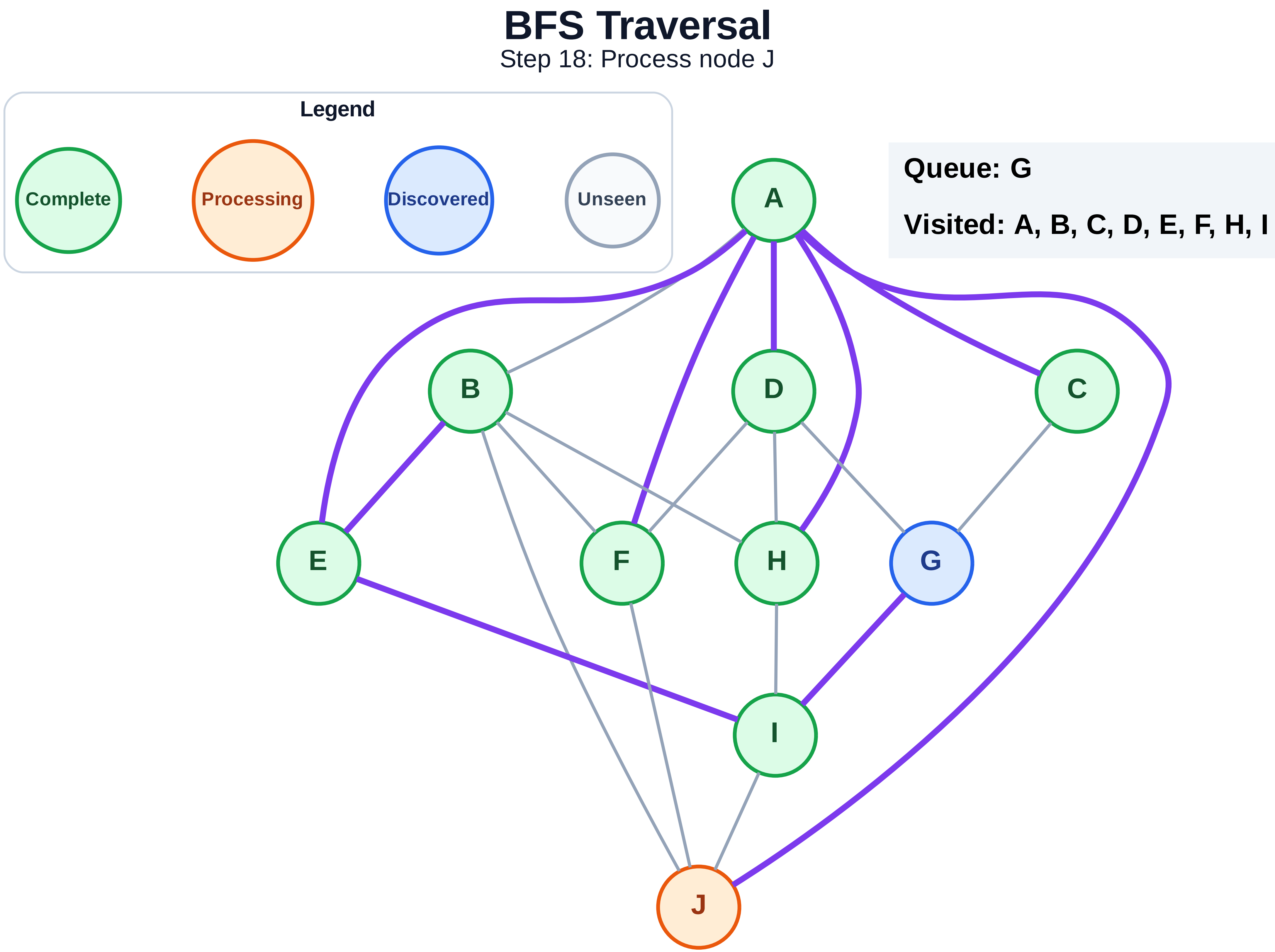
Step 16: Process node H



BFS Traversal

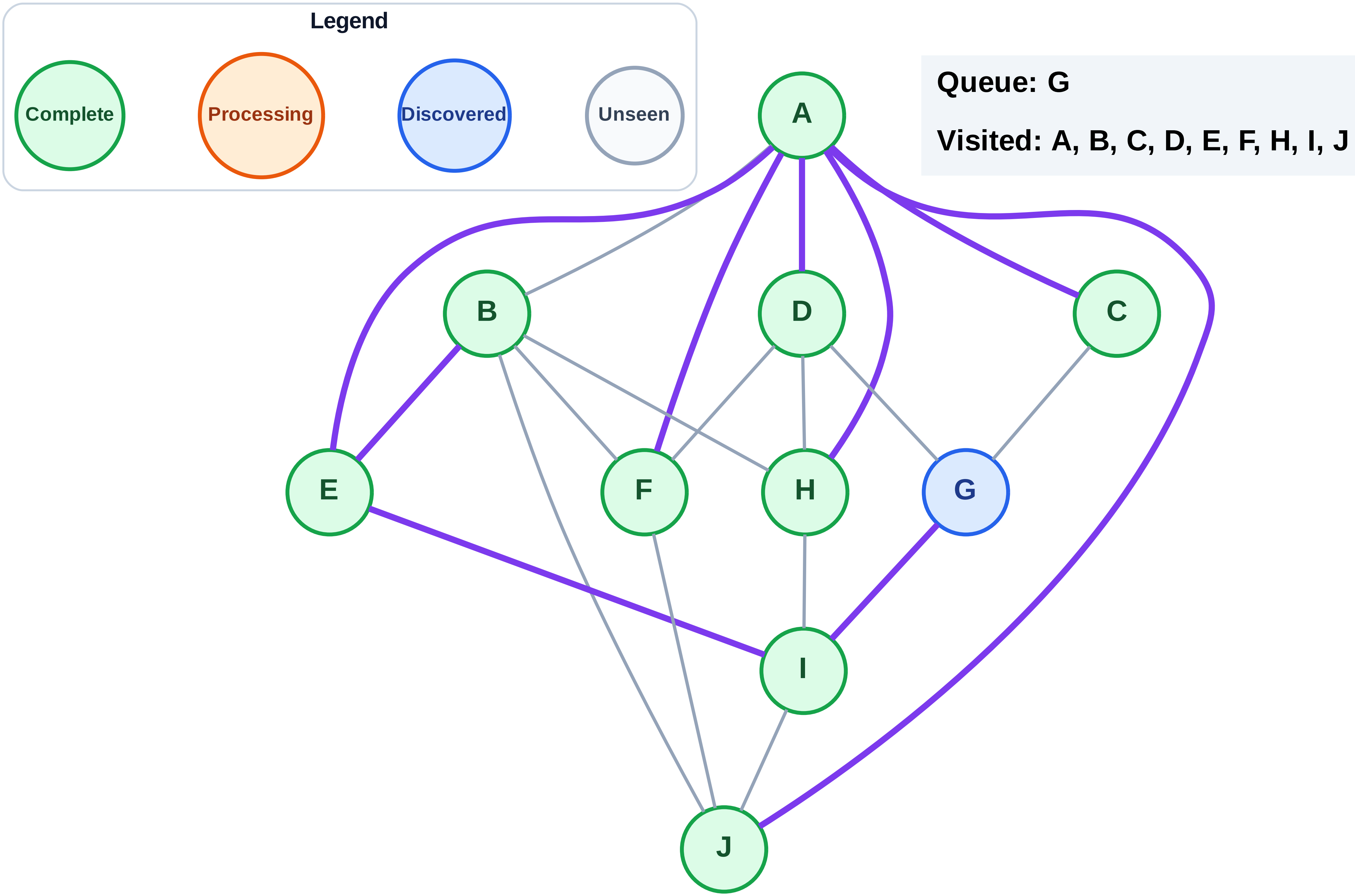
Step 17: No new neighbors from H

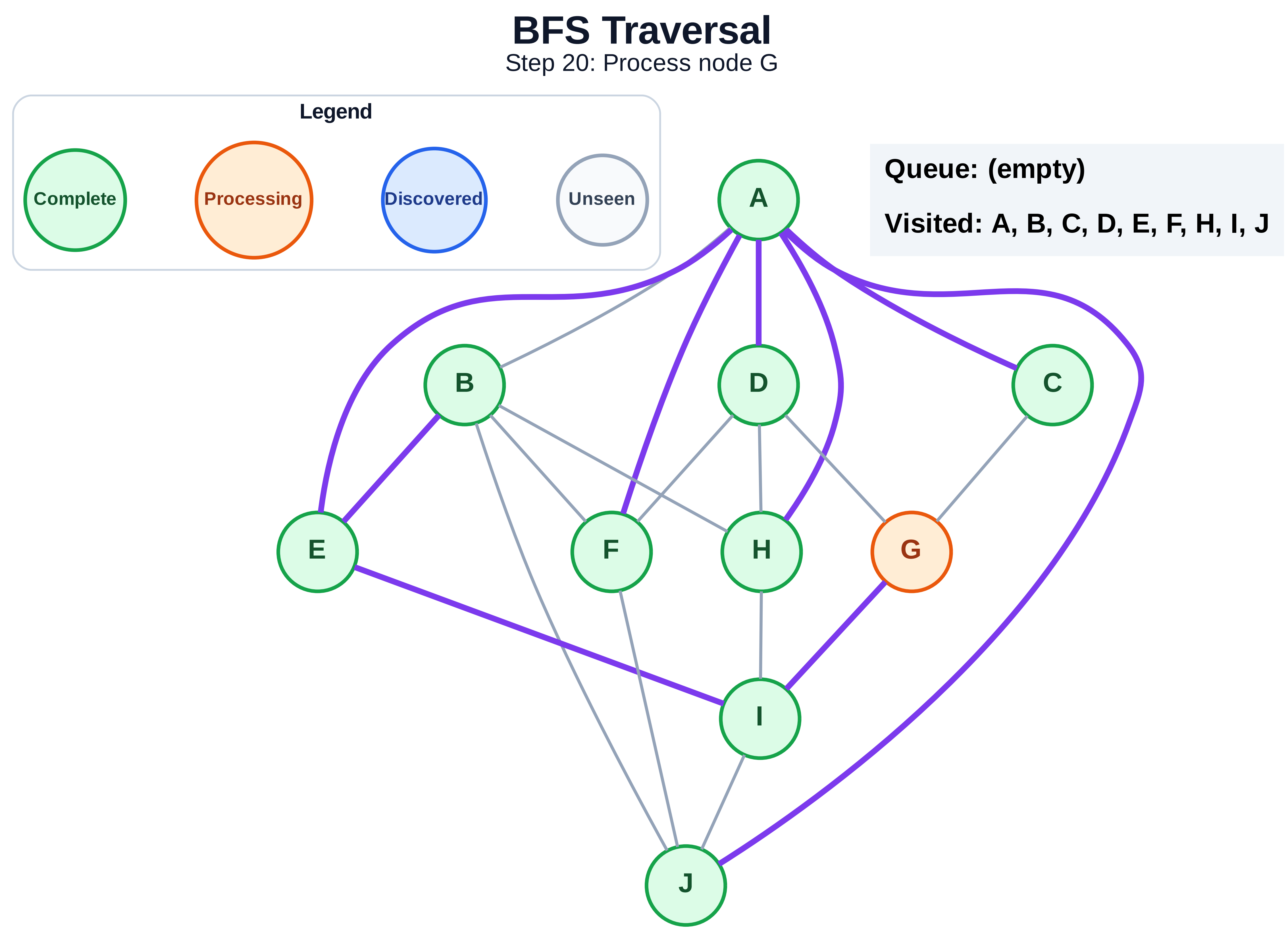



BFS Traversal
Step 18: Process node J
Legend
Complete
Processing
Discovered
Unseen
Queue: G
Visited: A, B, C, D, E, F, H, I
A
B
D
C
E
F
H
G
I
J

BFS Traversal

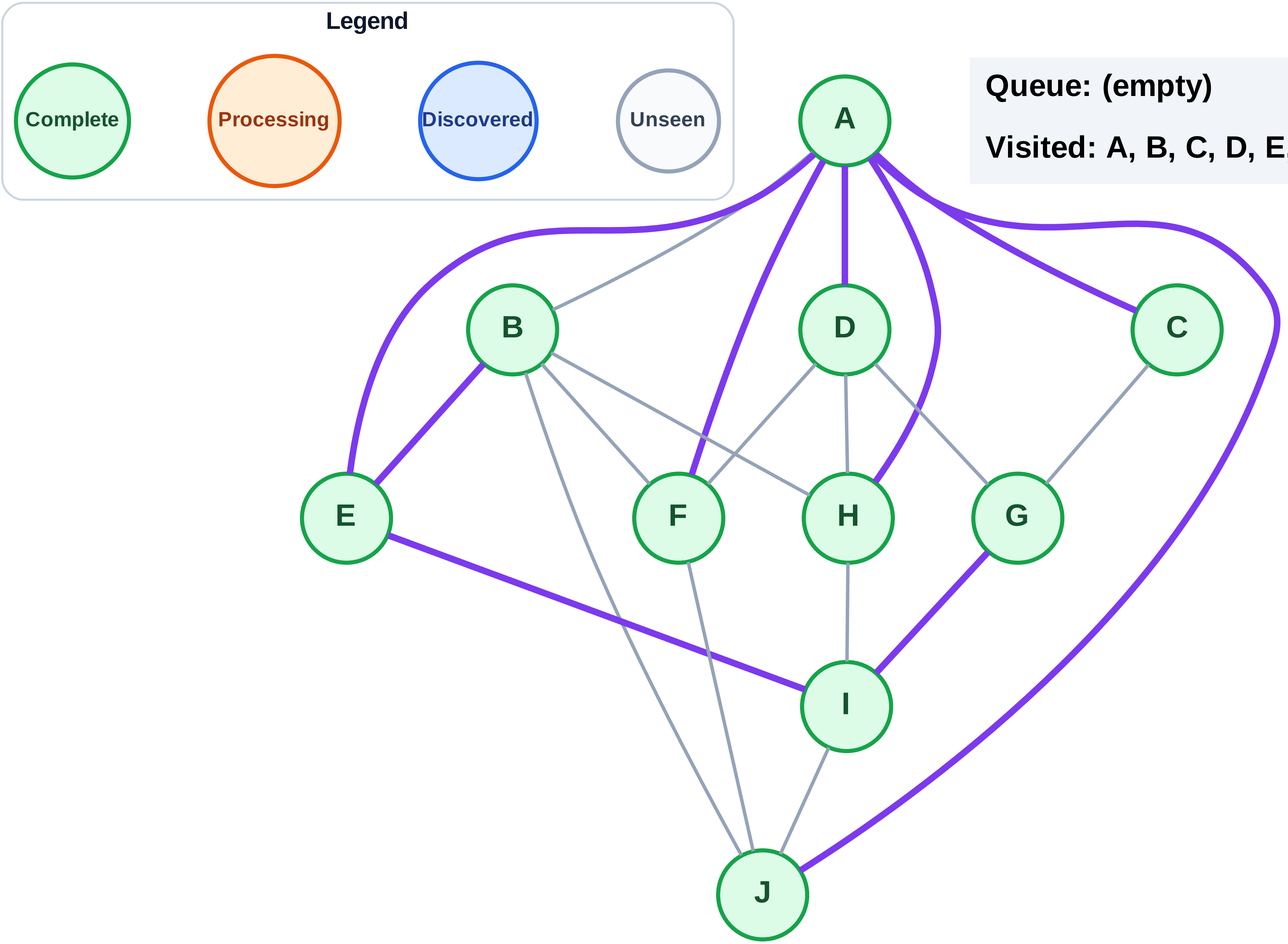
Step 19: No new neighbors from J




BFS Traversal
Step 20: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: A, B, C, D, E, F, H, I, J
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 21: No new neighbors from G

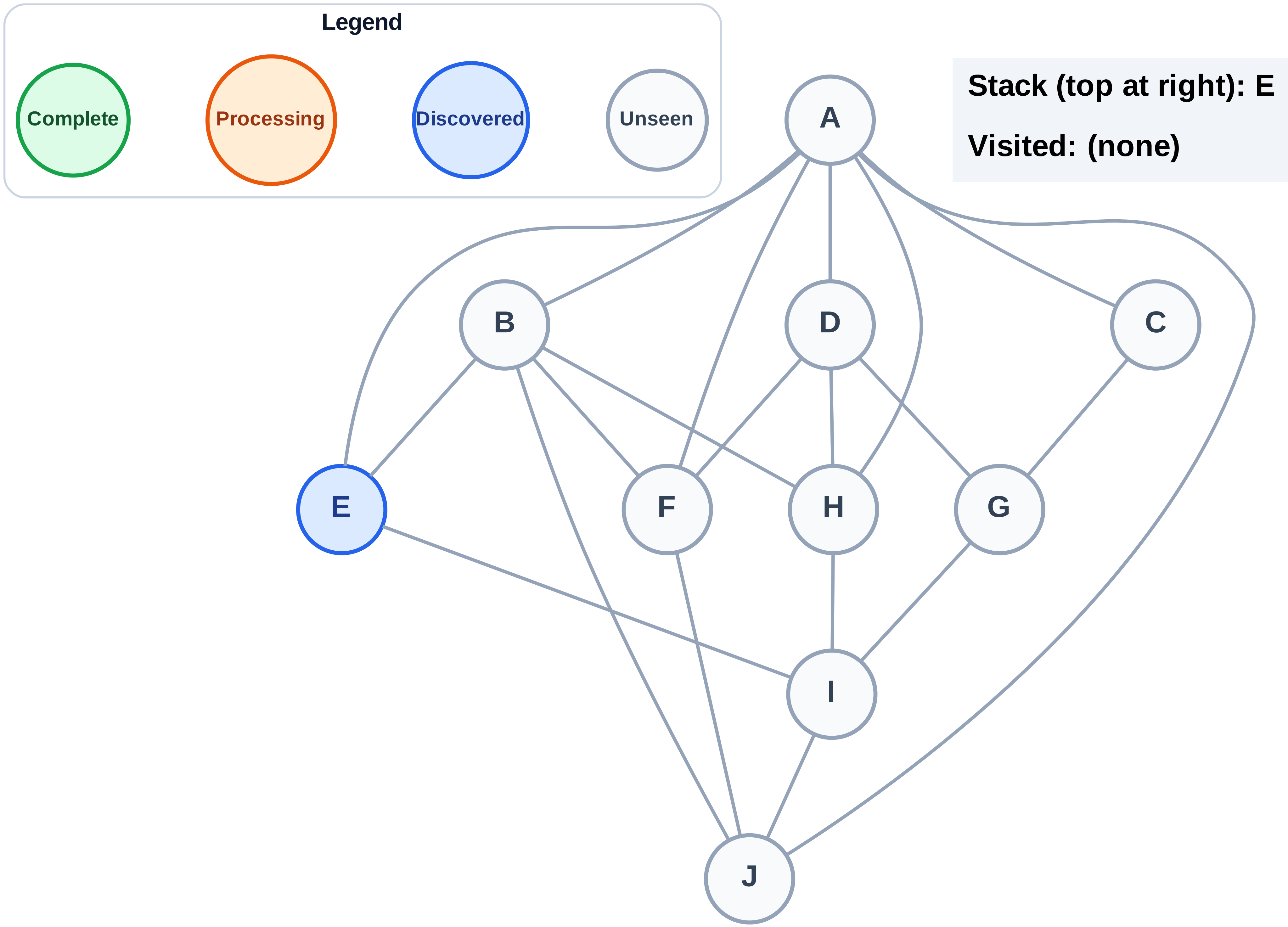


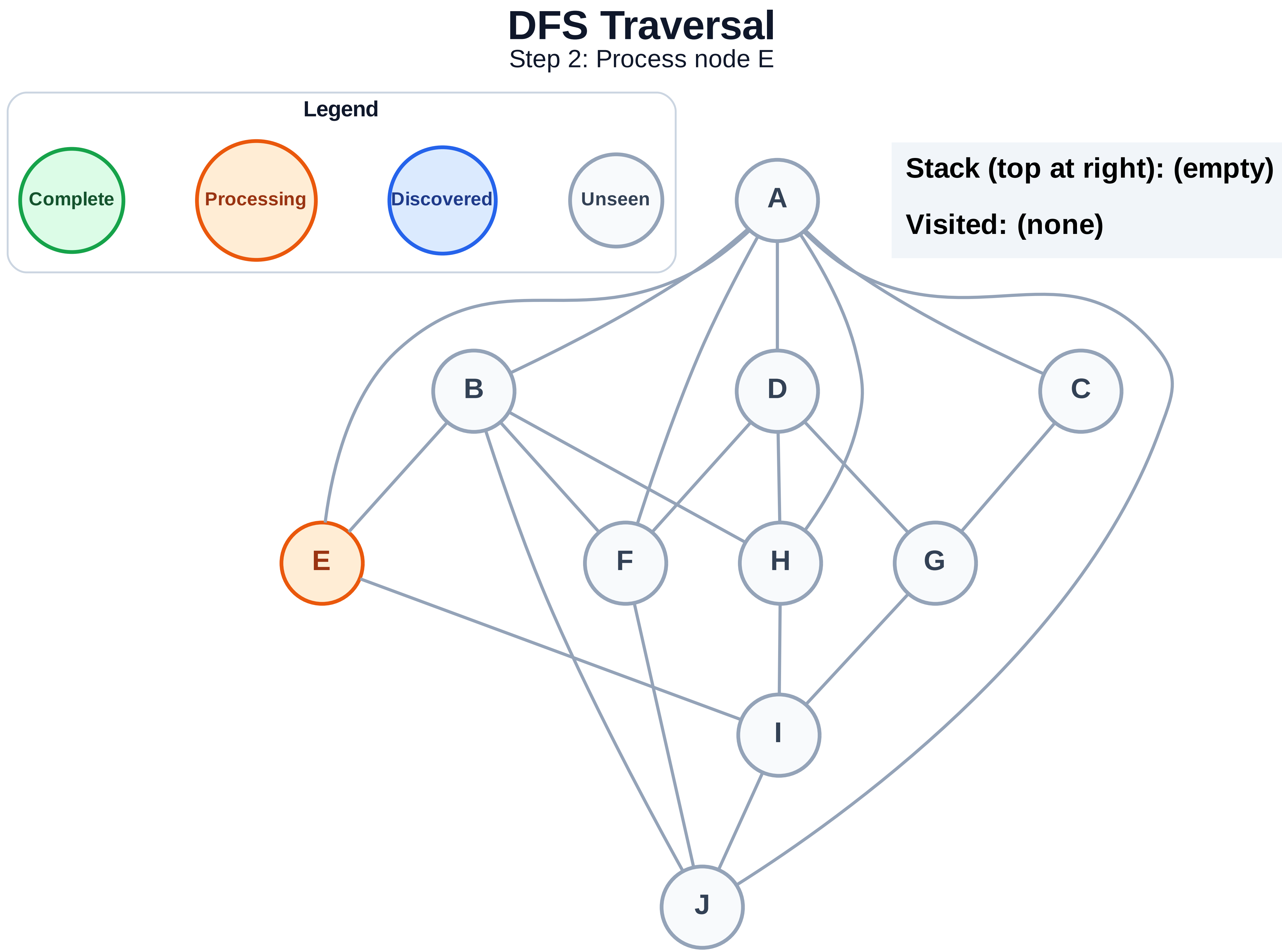
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at E

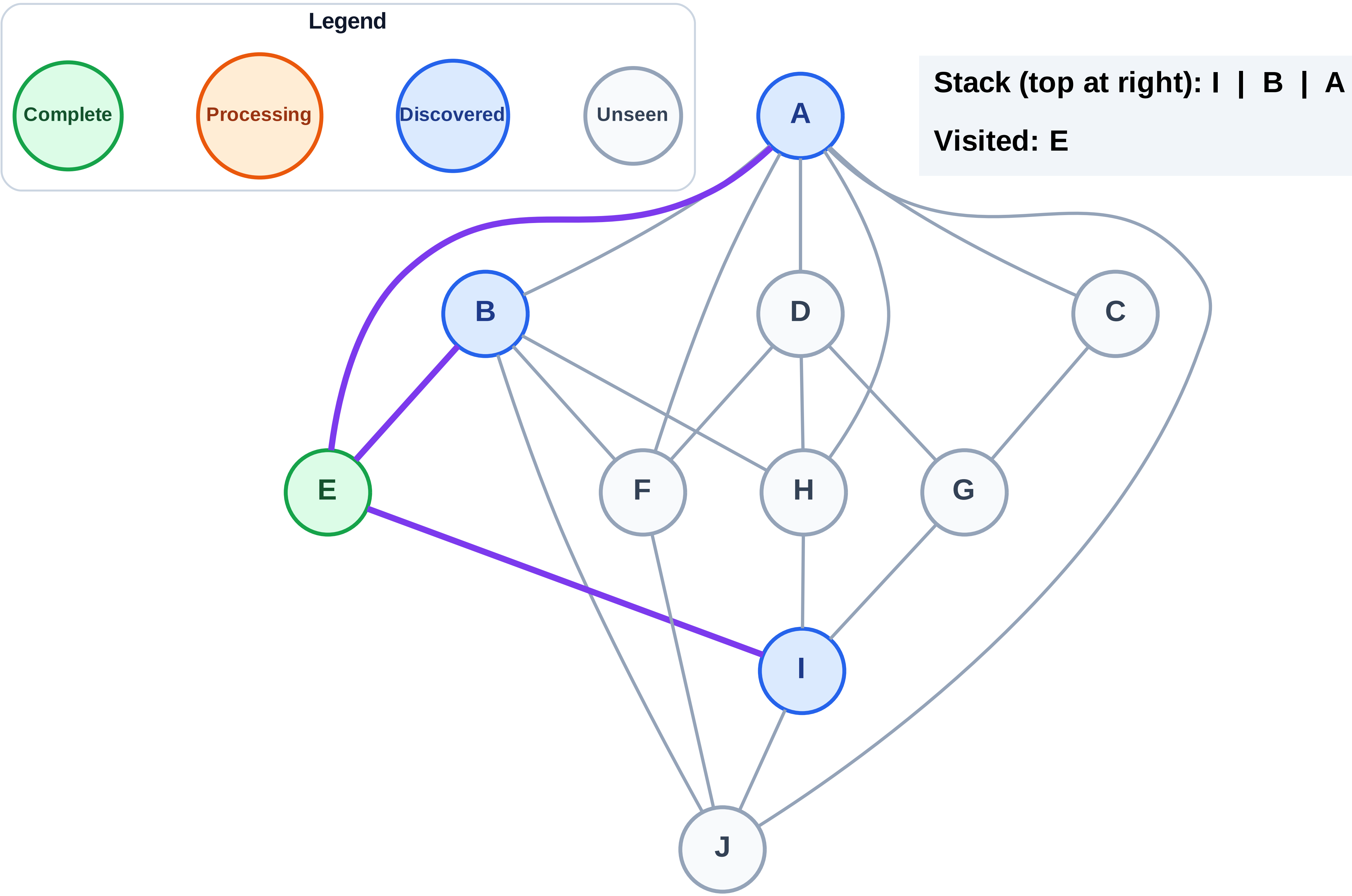




DFS Traversal
Step 2: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: (none)
A
B
D
C
E
F
H
G
I
J

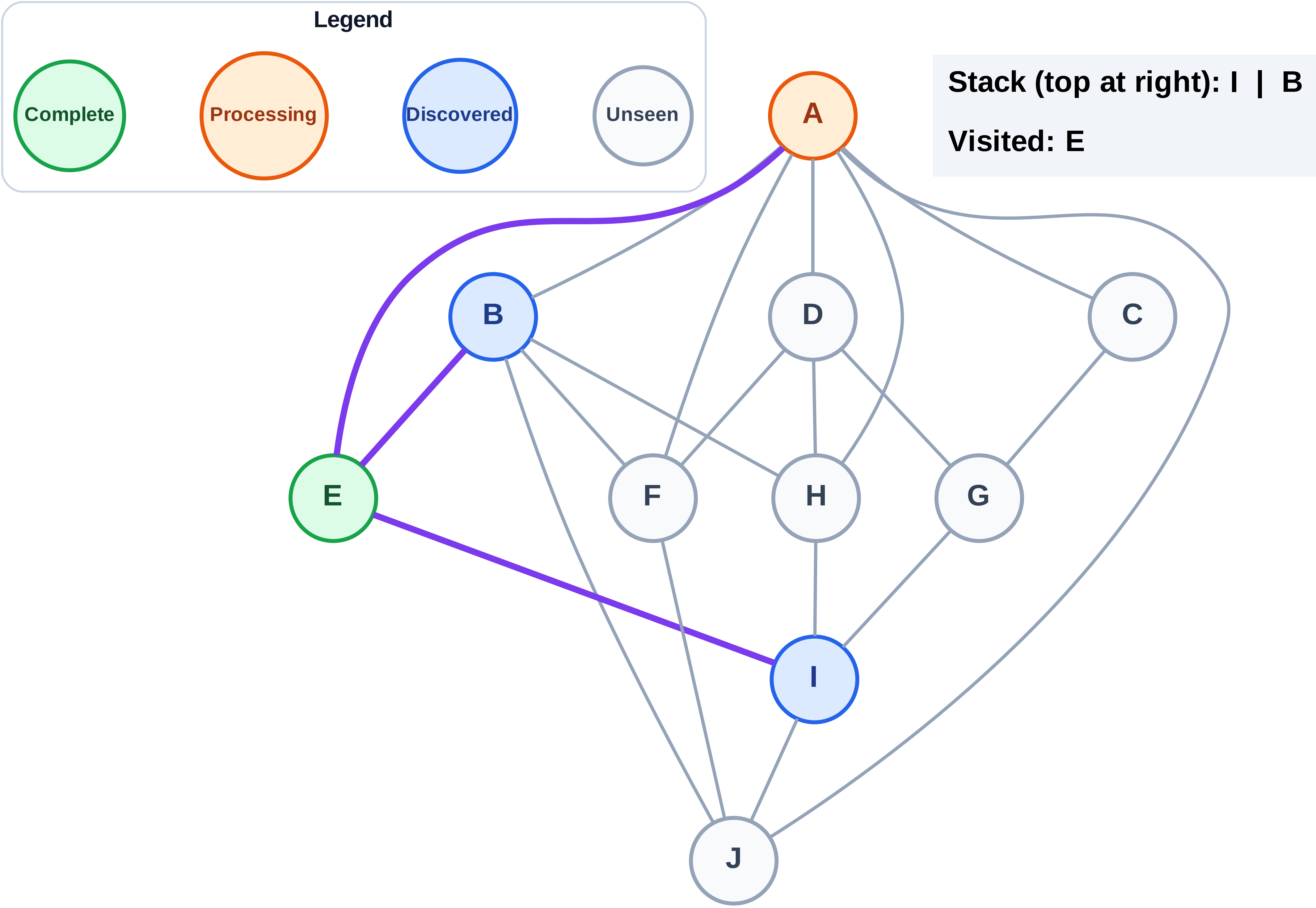
DFS Traversal

Step 3: Discover I, B, A from E



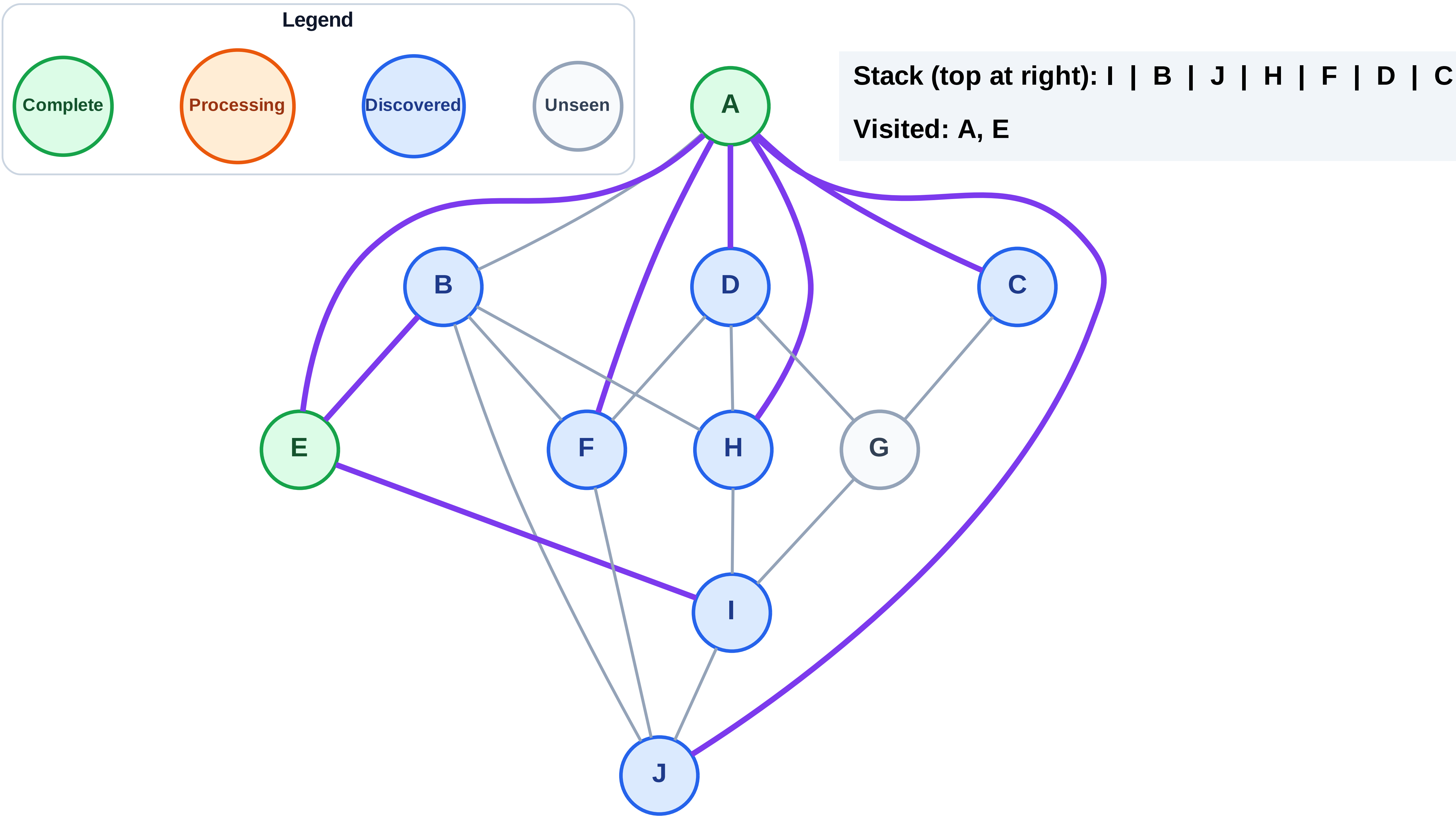
DFS Traversal

Step 4: Process node A



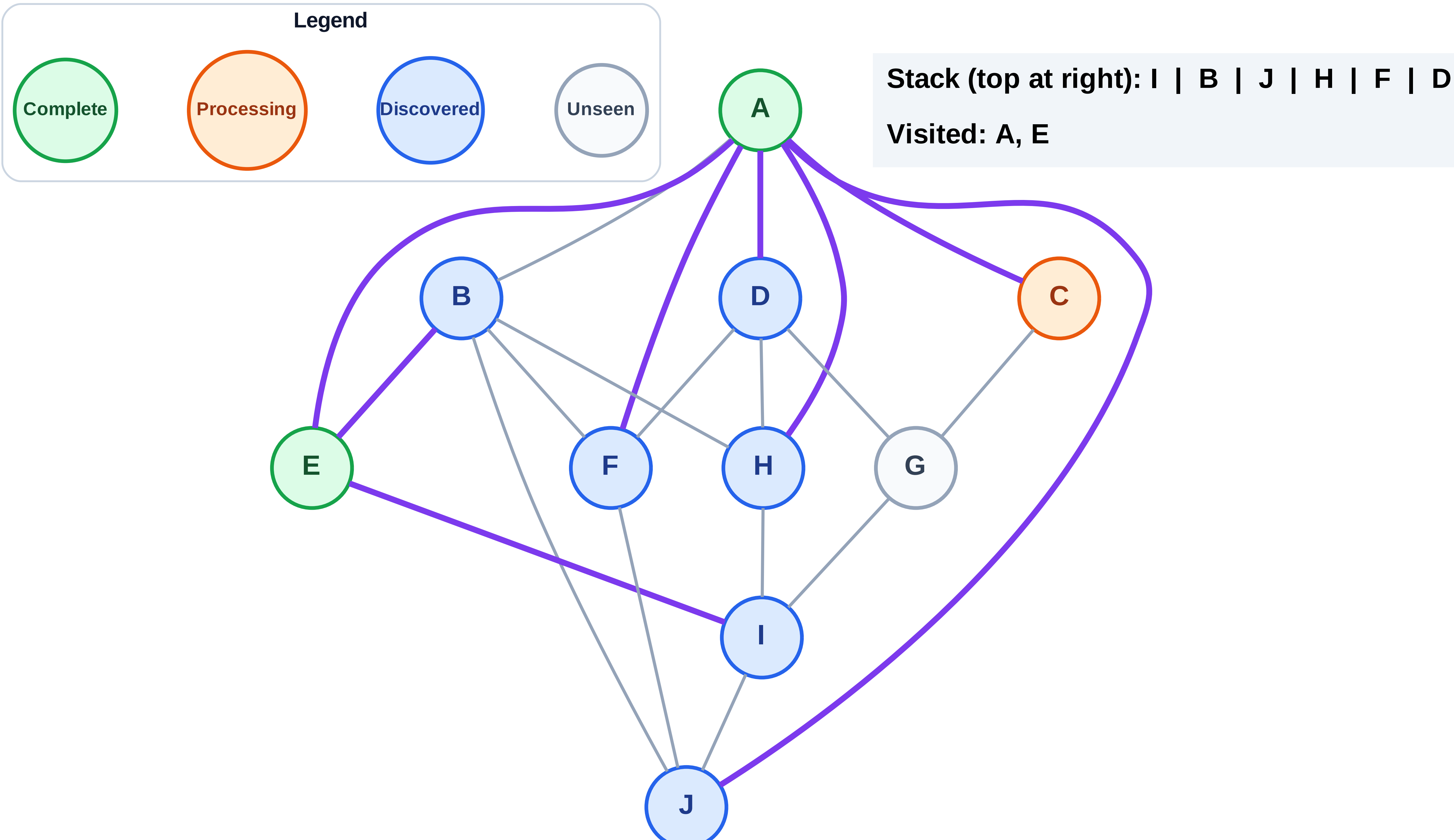
DFS Traversal

Step 5: Discover J, H, F, D, C from A



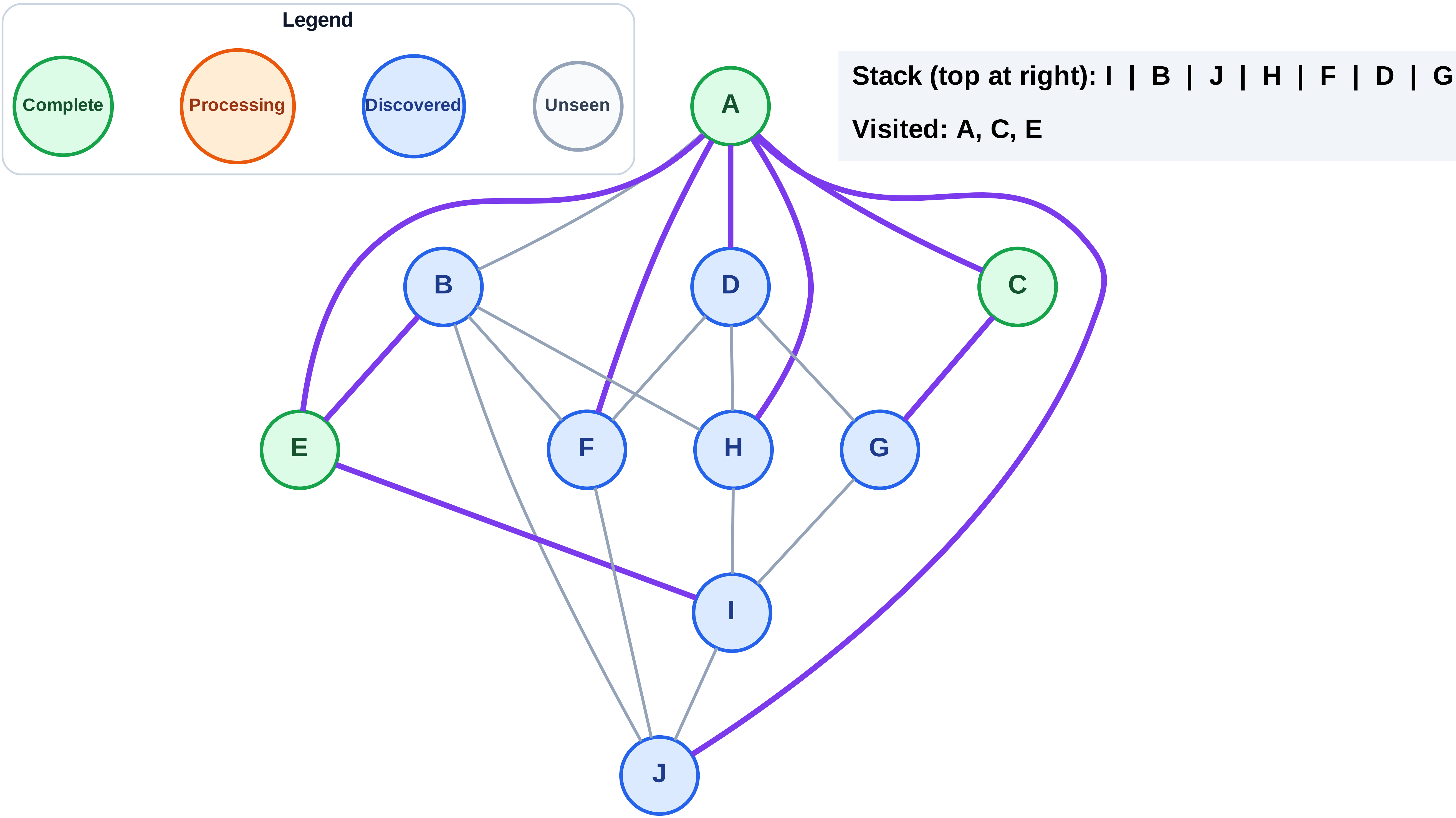
DFS Traversal

Step 6: Process node C



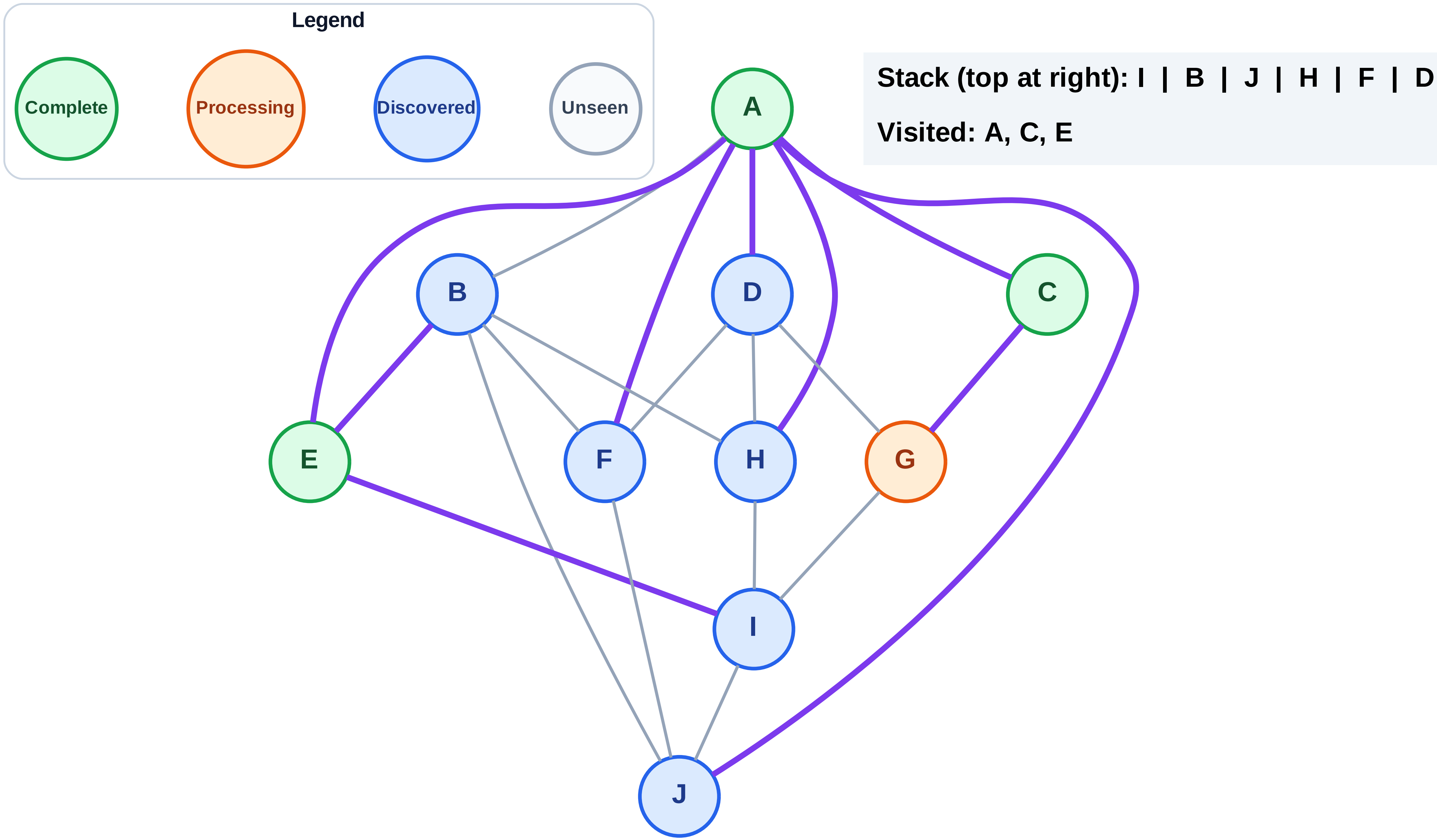
DFS Traversal

Step 7: Discover G from C



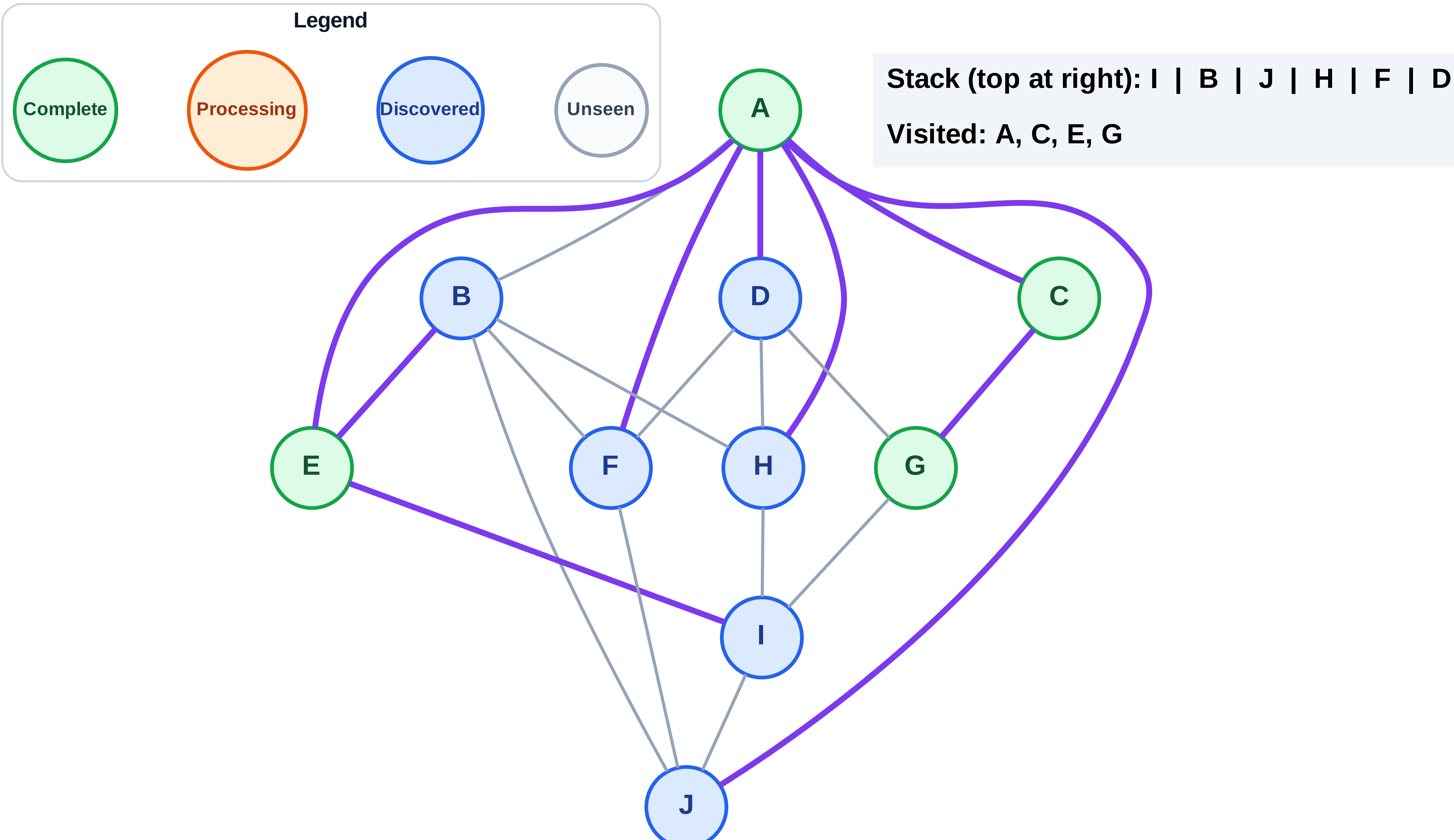
DFS Traversal

Step 8: Process node G



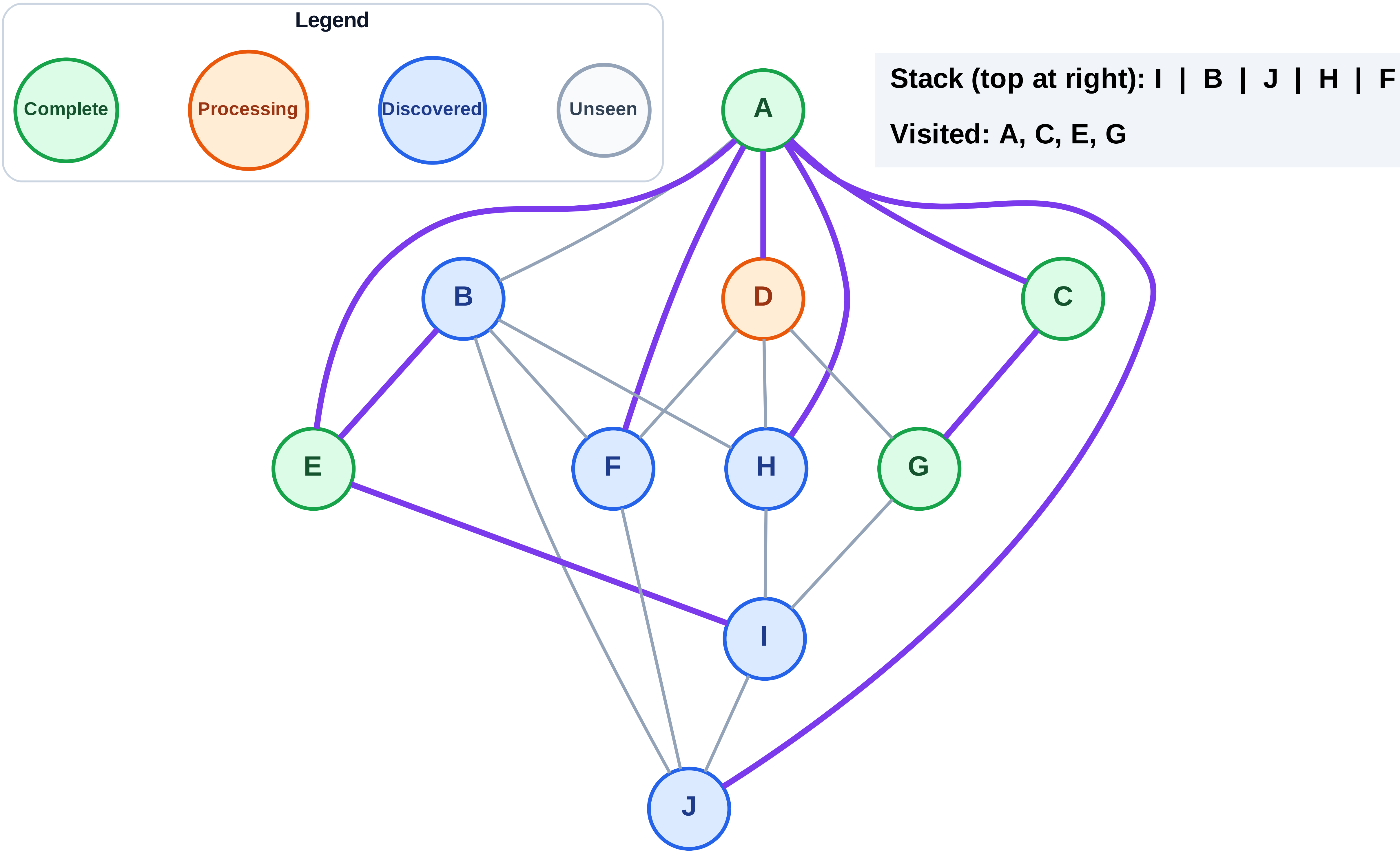
DFS Traversal

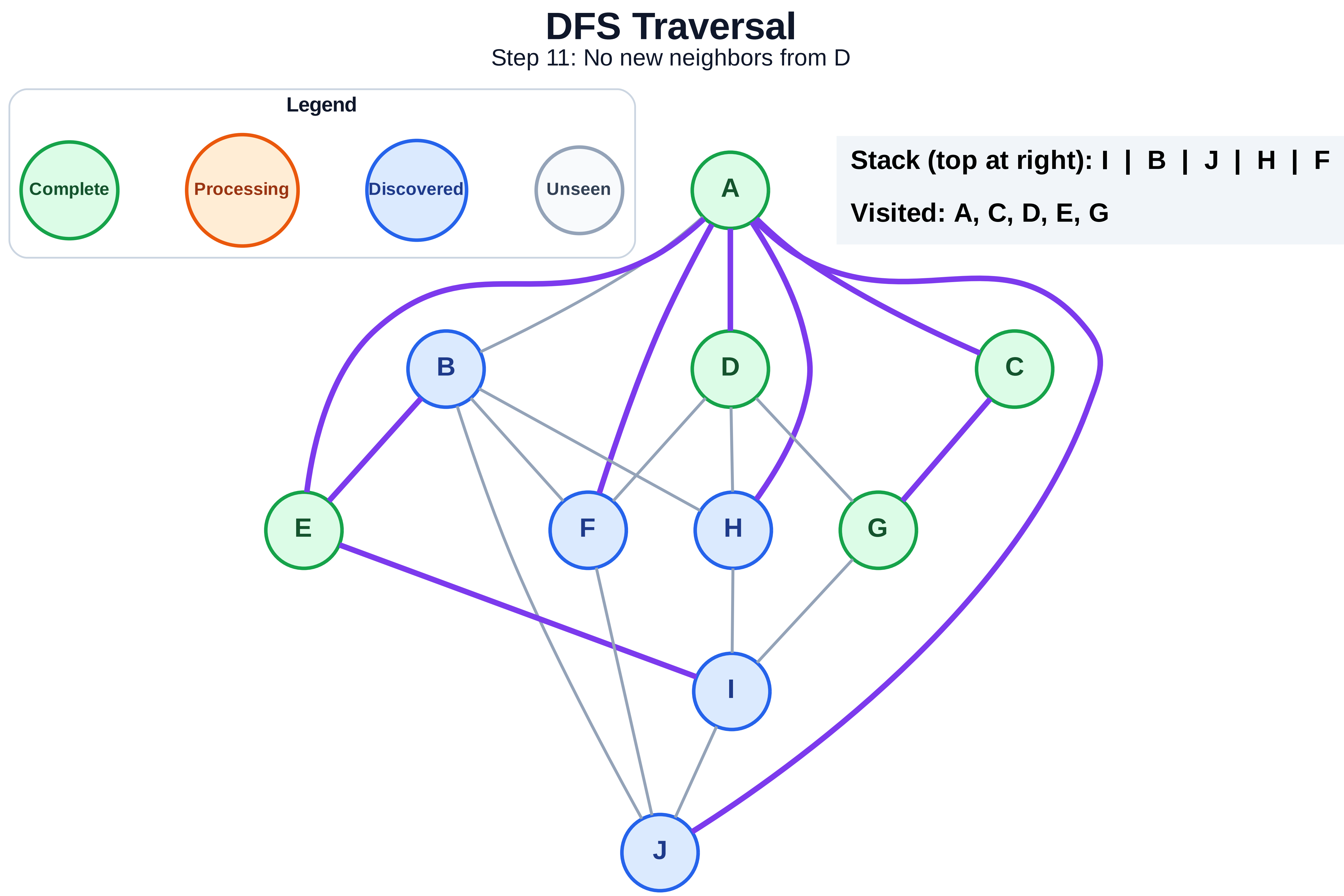
Step 9: No new neighbors from G



DFS Traversal

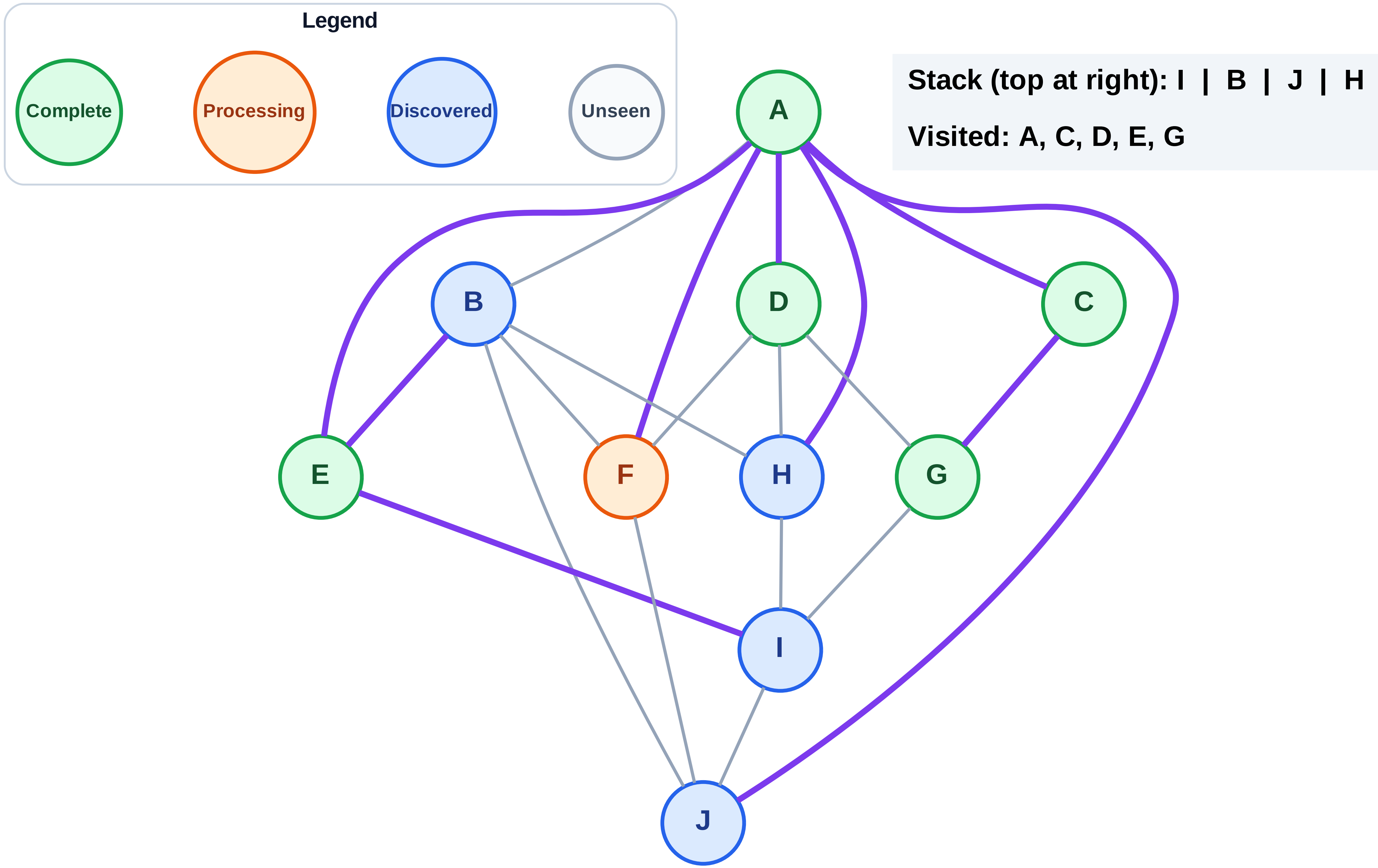
Step 10: Process node D

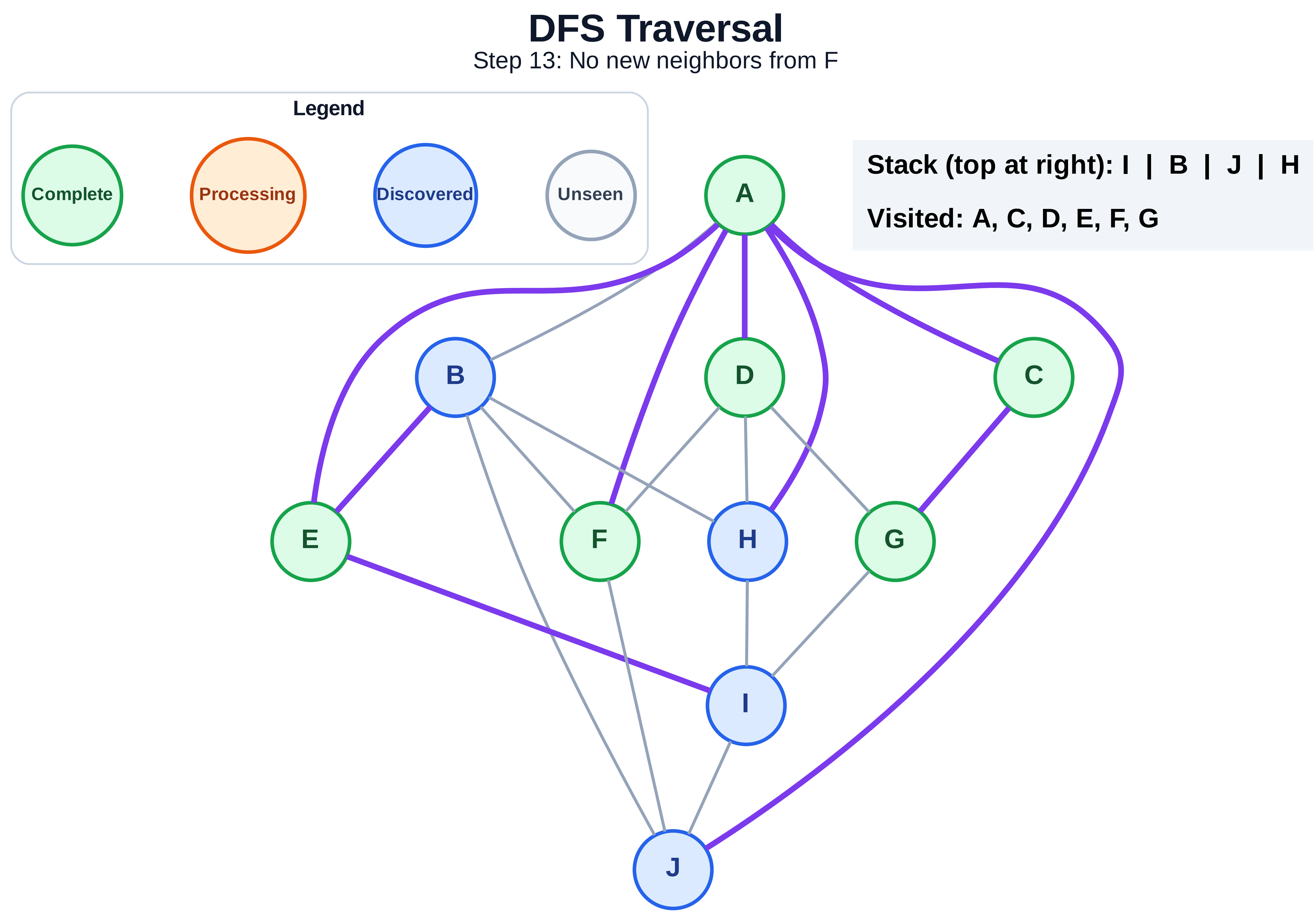



DFS Traversal
Step 11: No new neighbors from D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | B | J | H | F
Visited: A, C, D, E, G
A
B
D
C
E
F
H
G
I
J

DFS Traversal

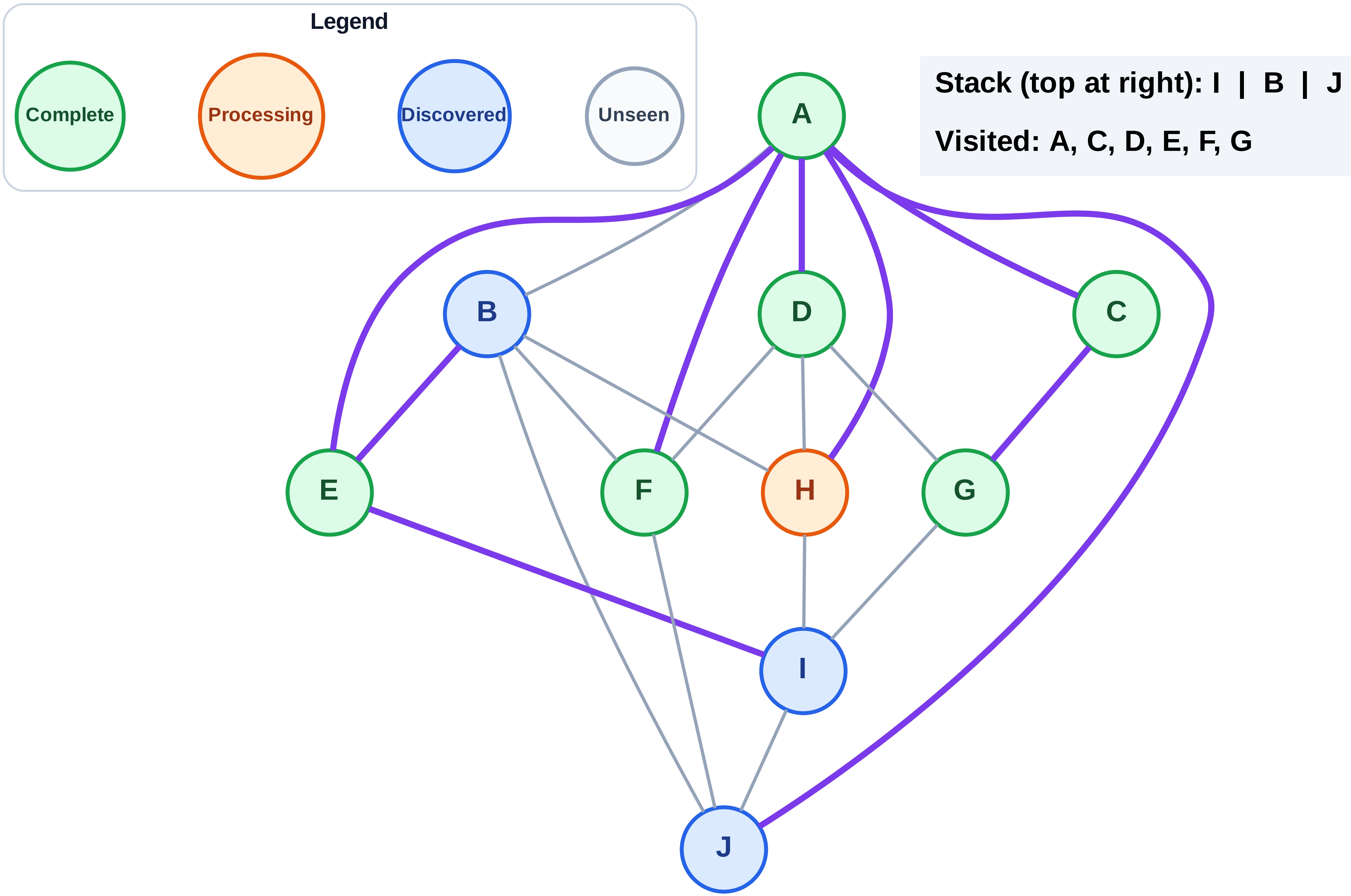
Step 12: Process node F




DFS Traversal
Step 13: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | B | J | H
Visited: A, C, D, E, F, G
A
B
D
C
E
F
H
G
I
J

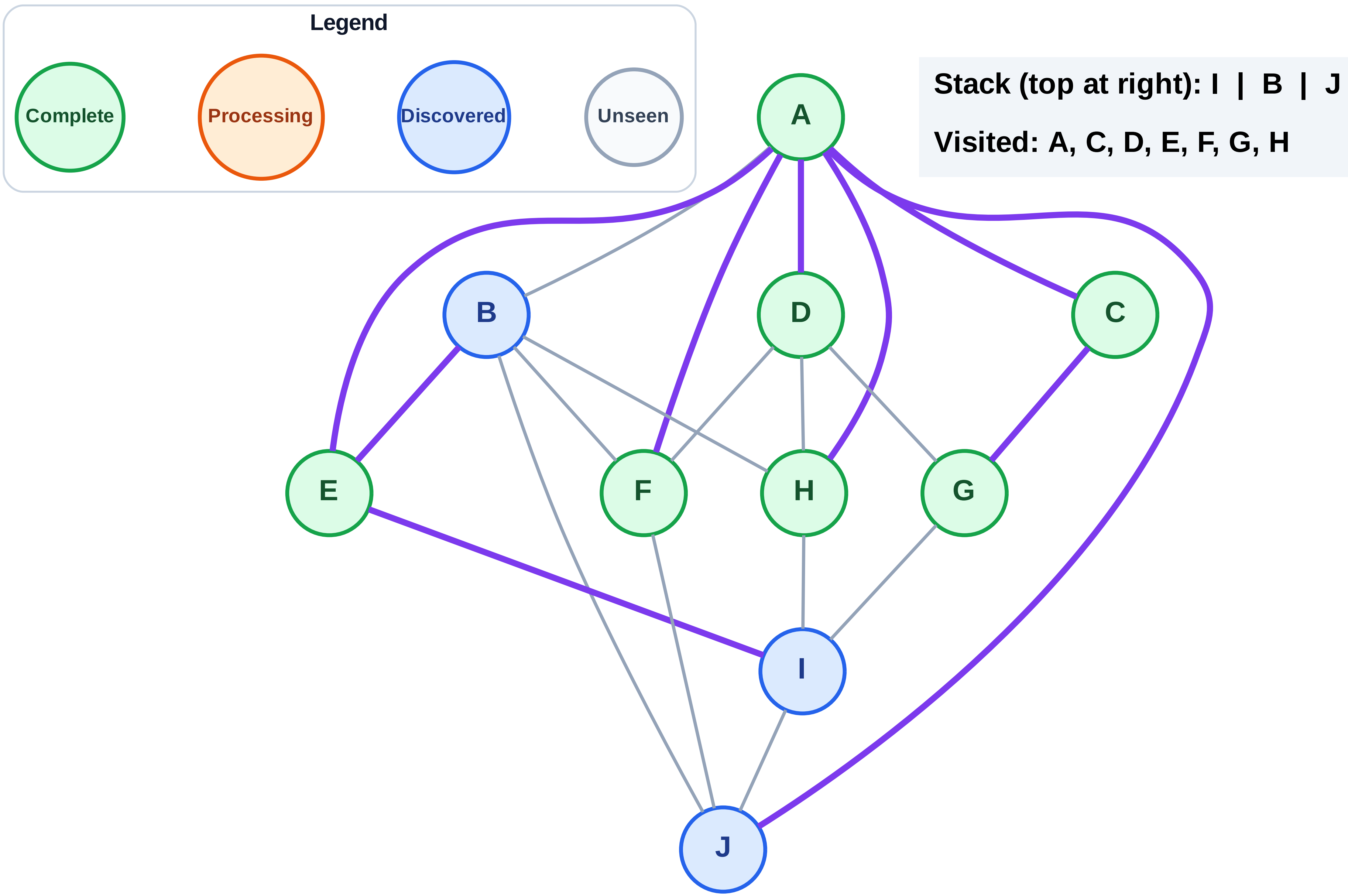
DFS Traversal

Step 14: Process node H



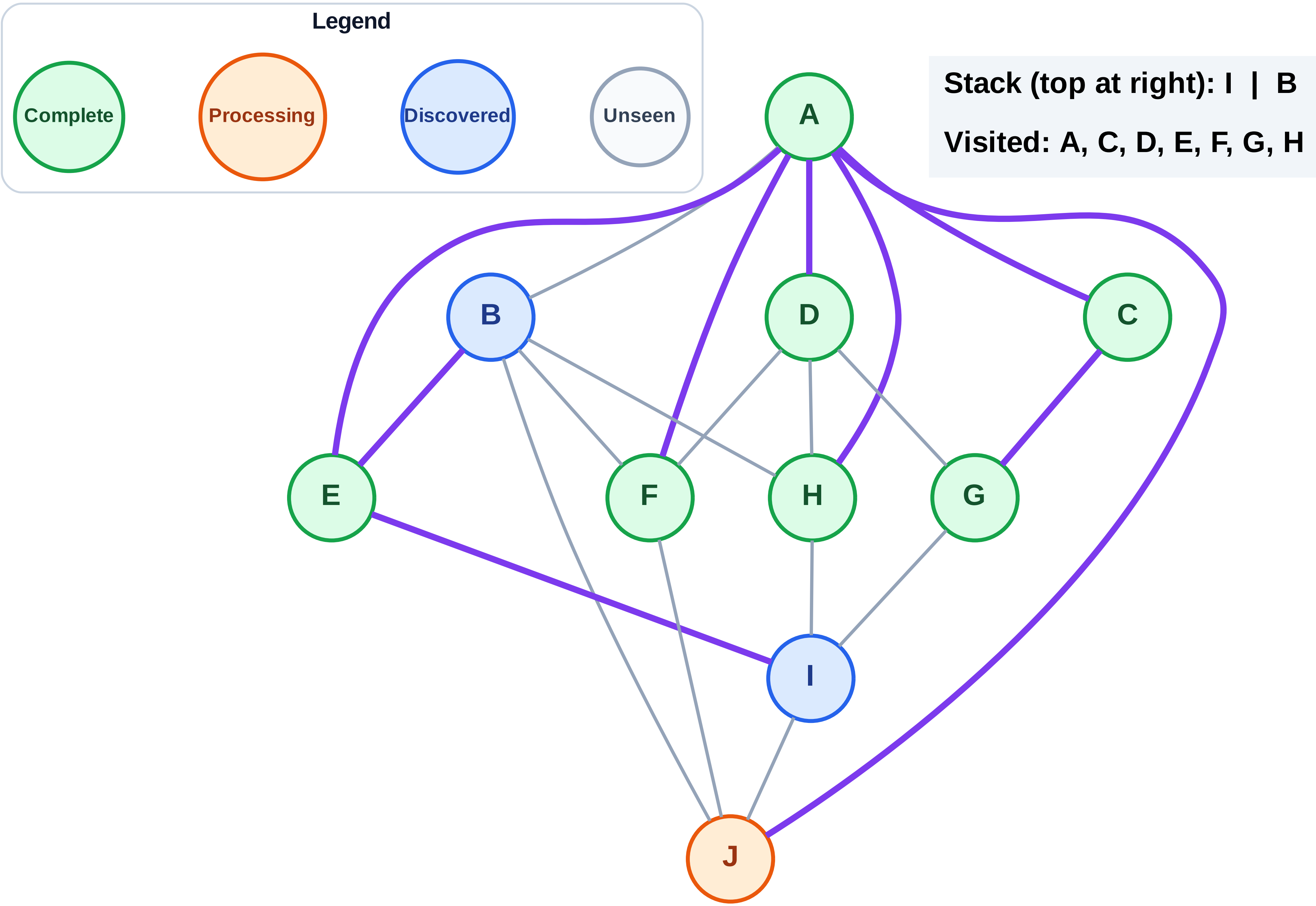
DFS Traversal

Step 15: No new neighbors from H



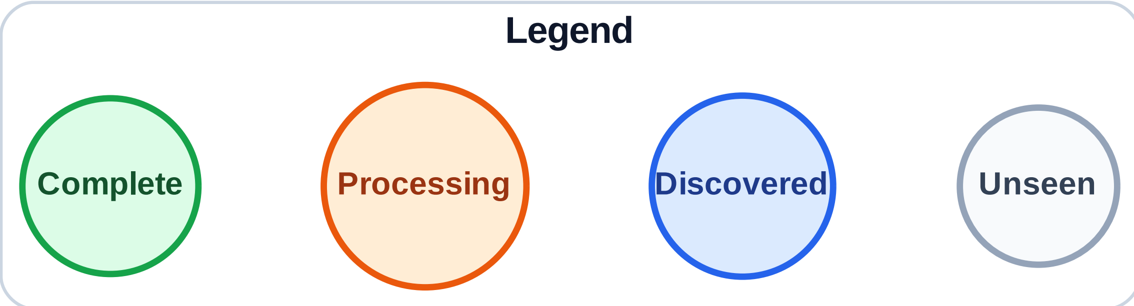
DFS Traversal

Step 16: Process node J



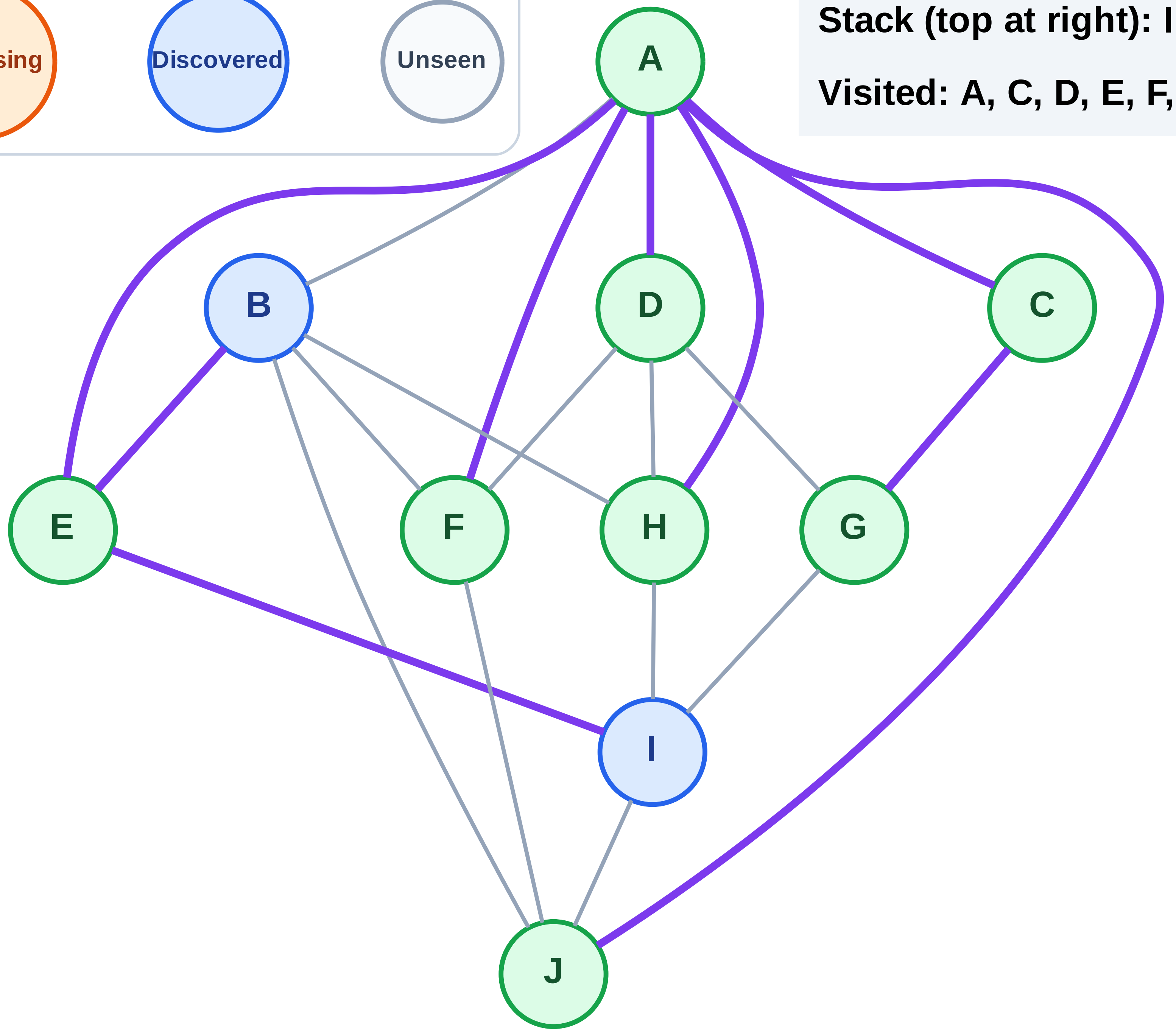
DFS Traversal

Step 17: No new neighbors from J



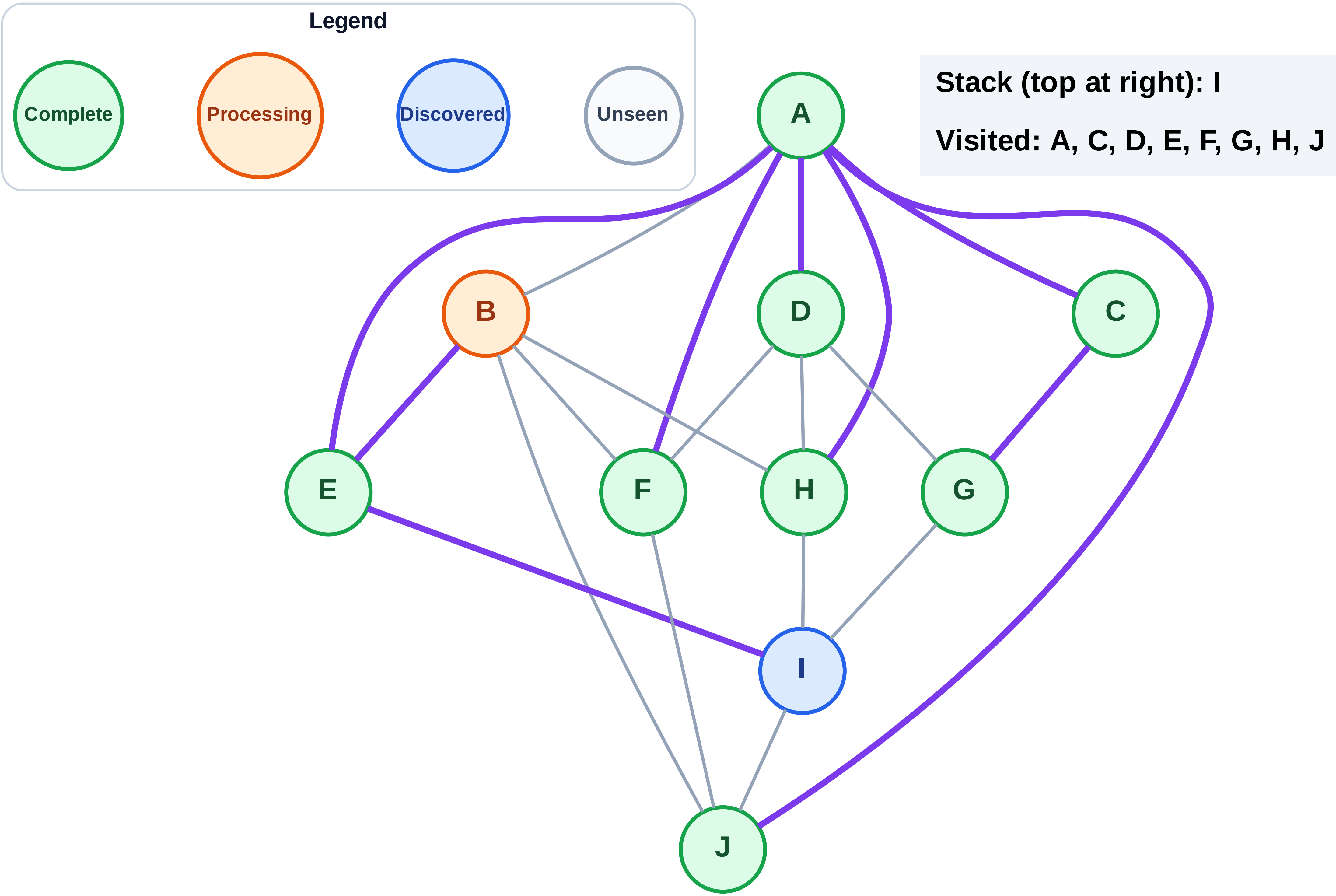
Stack (top at right): I | B

Visited: A, C, D, E, F, G, H, J



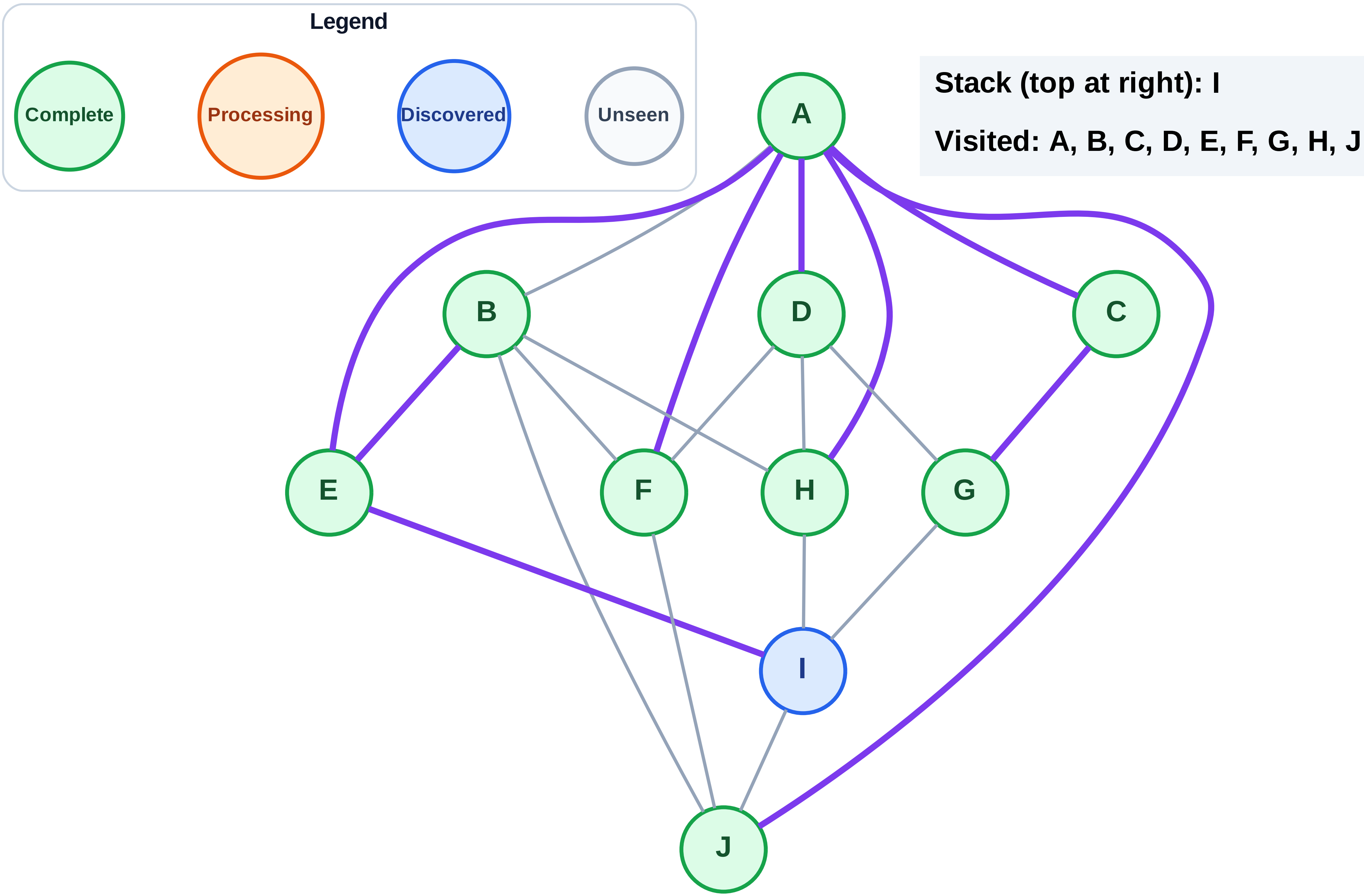
DFS Traversal

Step 18: Process node B



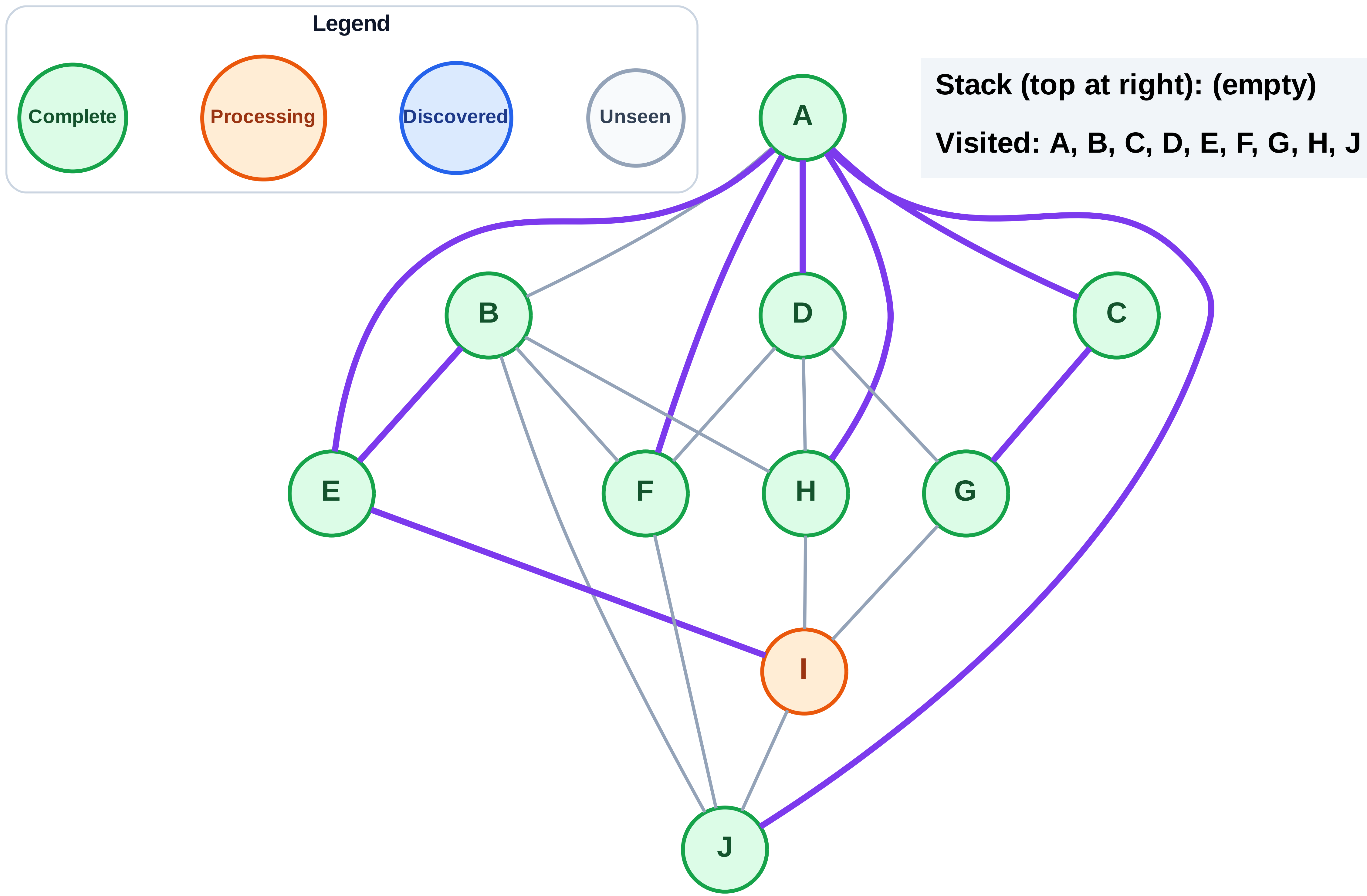
DFS Traversal

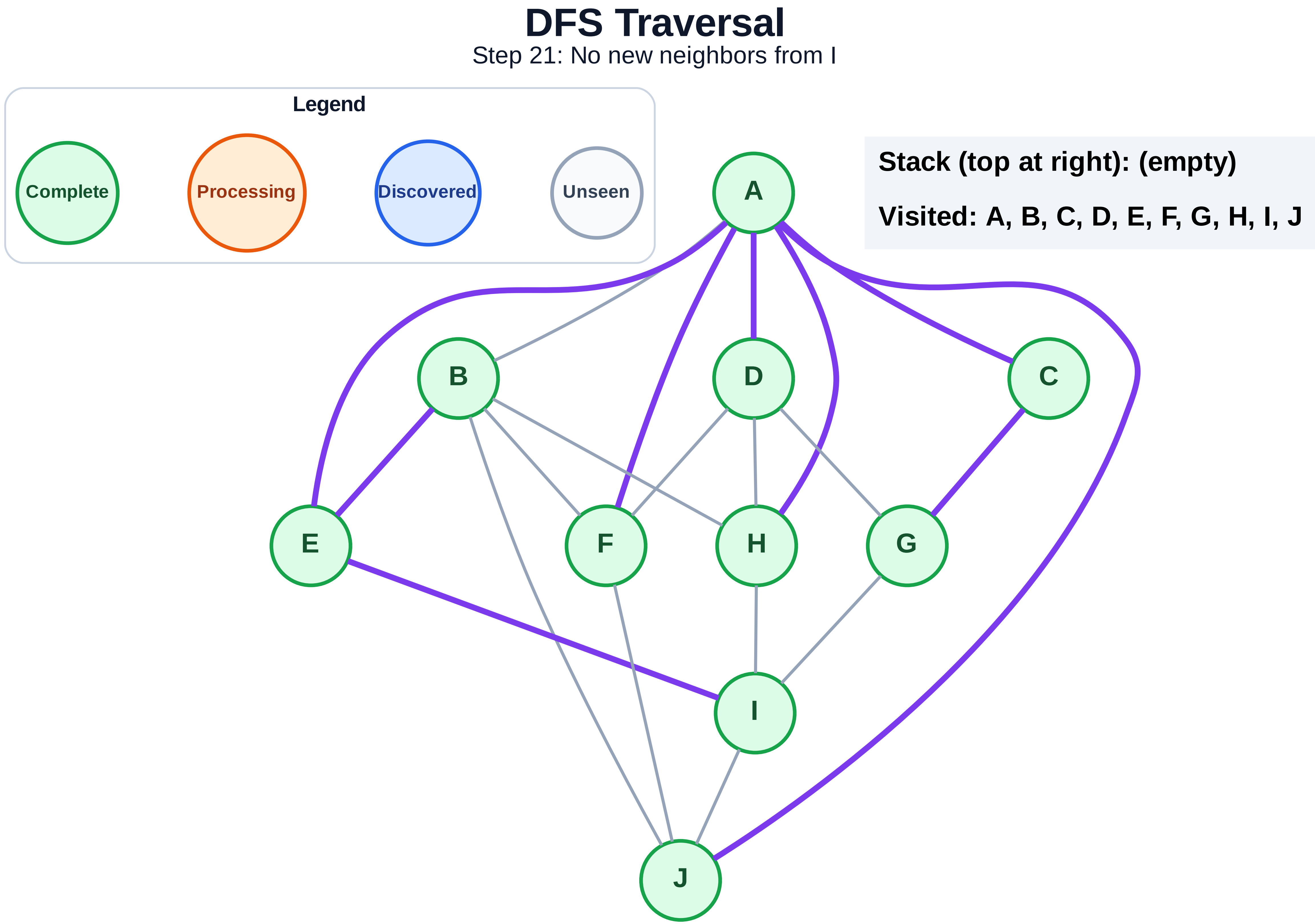
Step 19: No new neighbors from B



DFS Traversal

Step 20: Process node I




DFS Traversal
Step 21: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
D
C
E
F
H
G
I
J